

MOLTEN SALT REACTOR MICROSCOPIC DELAYED NEUTRON PRECURSOR GROUP
FORMULATION AND EVALUATION

BY

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DISSERTATION

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Abstract

Fluid-fueled molten salt reactors (MSRs) have garnered interest since the Generation IV International Forum in the early 2000s, where the MSR was designated as one of six advanced reactors of interest. A unique consideration in MSRs are delayed neutron precursors (DNPs), which move from areas of high importance in the core to areas of lower importance in the core and ex-core regions. This causes the effective delayed neutron fraction to decrease, leading to changes in reactivity based on the salt flow rate. DNPs are important for modeling and simulation of MSRs.

Many studies have investigated how the effective delayed neutron fraction changes based on the movement of fuel salt in MSRs. These studies do not account for the effect of fuel salt movement and reprocessing on the DNP group parameters used in the transient simulations. The effect of neglecting these MSR-specific phenomena is reduced accuracy in the DNP group parameters.

These phenomena are important to consider, as simulations use the DNP group parameters to enable safety, security, and safeguards by design. By using DNP group parameters with greater accuracy, models can more accurately reflect physical phenomena. Having greater accuracy in predictive models is key for developing an advanced reactor pipeline.

The DNP group parameters for MSRs are resolved using the microscopic, or aggregate or summation, method. This method entails using individual DNP data to reconstruct the DNP groups by summing them together in simulated irradiation experiments. Pulse irradiation experiments occur at very short time scales relative to MSR physical phenomena and are used to generate the short lived DNP group parameters. Saturation, or infinite, irradiation experiments occur at time scales several times longer than the longest DNP half life and are used to generate the longer lived DNP group parameters. Because the saturation irradiation lasts on the order of minutes, the thermal-hydraulic and chemical effects in MSRs have a non-negligible impact. Therefore, the longer lived DNP group parameters will vary, while the shorter lived DNP group parameters will remain the same.

A non-linear least-squares solver is used to generate a χ^2 minimized set of DNP group parameters. A specific DNP group parameter set is proposed for a particular set of conditions in the Molten Salt Reactor Experiment (MSRE). Verification is conducted via comparison with total delayed neutron yields and average half-lives from the literature. Validation is conducted via comparison with the MSRE during pump start-up and coast-down.

The results demonstrate the validity of these MSR phenomena-inclusive DNP group parameters. The methodology used to generate these parameters can be replicated for other MSR conditions, enabling the formation of more accurate sets of DNP group parameters for MSRs. This improves the accuracy of MSR simulations, which is important for safety, security, and safeguards by design as well as for developing an advanced reactor pipeline. Proposed work includes the following: development and evaluation of a more sophisticated model for generating the DNP group parameters, generation of polynomial fits for various parameters used in the DNP group generation, generation of DNP group spectra, uncertainty quantification, analysis of the flux shape effect on DNP group parameters, and evaluation of various datasets on DNP group parameters.

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Chapter 1

Introduction

- Go through history of group parameters (maybe timeline figure?).
- Discuss MSRs.
- "The next step forward for group parameter development is to have group parameters that accurately reflect the physics that occur in fluid-fueled molten salt reactors (MSRs)".
- Introduce the gap [151, 155, 78, 132, 72] that other works do not account for (they investigate movement of DNPs during transient, not for delayed neutron precursor (DNP) group generation).
- Discuss methodology at a high level.
- Discuss results at a high level.
- Discuss conclusions of the work at a high level.

Overview of rest of document

1939 Roberts et al. discovery of delayed neutrons [115]. 1946 initial spectral measurements [24]. In 1947, Hughes et al. initial use of six groups (kinda) at Argonne heavy water pile [59]. In 1957, Keepin used a sample of U235 in Godiva to generate DNP group parameters (improved group fit). Thus far, only macroscopic approaches have been possible due to insufficient data. For MSRs which have chemical removal of certain elements, tracking individual nuclides is important. In 1989, Brady's thesis used updated data on 271 nuclides identified as DNPs (first attempt at microscopic/summation/aggregate approach). This data was used for ENDF-B/VI was released, and since then there have not been any changes to the DNP group parameters [146, 23]. In 1999, Parish says "Based on the six-group data reported in the literature, it appears that the ENDF/B-VI six-group constants are in need of revision." [99]. In 2014, International Atomic Energy Agency (IAEA) Coordinated Research Project (CRP) on Development of a Reference Database for β -delayed neutron emission data [92]. 2020 Leconte uses more than 400 DNPs with data [77].

Brady's thesis has a good history (up to 1988).

This document should contain tabulated data for various fissile nuclides, neutron irradiation energies, in-core residence times, ex-core residence times, and mixing rates. Additionally, data should be provided on the exact concentrations of each nuclide (up to 99% of the DN yield).

Could also discuss various datasets. [87] - shows various datasets, most important nuclides, and the error associated with them (2012). [62] - Lots of delnu data (2014).

Chapter 2

Background and Motivation

This work aims to generate more accurate DNP group parameters for MSRs and implement those parameters for safety, security, and safeguards design purposes. These DNP group parameters in MSRs differ because fuel circulates and undergoes online chemical reprocessing. These effects can be captured in spatially resolved MSR fuel depletion simulations. The existing methodology for DNP group parameter generation does not account for the physical phenomena present in MSR fuel depletion [66, 137, 86, 13, 145, 21, 59, 84, 85, 106, 32, 106, 123, 146, 20, 77]. More accurate DNP group parameters are generated by accounting for these physical phenomena, which is particularly important for safety, security, and safeguards related concerns.

2.1 Delayed Neutron Precursors

Before delving deeper into the DNP group parameters, it is useful to first understand DNPs conceptually. DNPs are an important consideration in nuclear reactors [33]. DNPs are nuclides which decay via β^- decay, producing a daughter, called an emitter, which emits a neutron [10]. In this work, the term DNP will be solely used rather than referring to emitters, as the delay of the neutron emission is effectively only due to the β decay [18, 98]. Classifying between the emitter and the DNP is not relevant for this work.

This delay in neutron emission significantly affects the reactor period and transient behavior. Equation (2.1) shows the time-dependent rate of delayed neutron emission from a single DNP as:

$$\dot{n}_{d,i}(t) = P_{n,i} \lambda_i N_i(t), \quad (2.1)$$

where,

$\dot{n}_{d,i}$ = the delayed neutron activity, or count rate, from DNP i [$\frac{n}{s}$]

$P_{n,i}$ = the average number of delayed neutrons emitted from
nuclide i , or the emission probability [-]

λ_i = the decay constant of DNP i [s^{-1}]

$N_i(t)$ = the number of DNP i [*atoms*].

In MSRs, the movement of the fuel salt necessitates spatial resolution for depletion, transmutation, and neutron emission. In a solid fuel reactor, such as a light water reactor (LWR), spatial resolution is not necessary to properly model the DNPs, as the fuel remains stationary. Modeling space and time together largely increases computational cost, making problems involving DNPs in MSRs more challenging than handling DNPs in stationary fuels.

Properly modeling the DNPs in reactors is important, otherwise simulations may predict incorrect power responses, reactivity worth, and reactor controllability [33, 34, 73]. Section 2.1.6 shows how the power response and reactivity worth depend on the DNPs. The reactor controllability varies based on the available number of DNPs. If there are fewer delayed neutrons available due to movement of the fuel salt, such as in a MSR, then the reactor period becomes shorter and makes the reactor harder to control.

2.1.1 Measurement

Macroscopic and microscopic approaches are used for measuring DNPs [99, 123]. The macroscopic approach uses a fissile sample irradiation to get the delayed neutron count rate, which can be approximated using groups. Conversely, the microscopic approach uses the data from many experiments combined together to simulate a fissile sample irradiation and generate the delayed neutron count rate. The microscopic approach is more challenging, as it depends on the work of many experimentalists. However, it offers more detail on the delayed neutron count rate.

Macroscopic Measurement of Delayed Neutron Precursors

This section covers the macroscopic approach. Analysis and collection of macroscopic experiment data is common in the literature [66, 137, 86, 13, 145, 21, 59, 84, 85, 30]. The macroscopic approach entails irradiating a sample containing a single fissile nuclide at or near a given neutron energy. Small samples are used to minimize neutron multiplication within the sample. Two different types of irradiation are used: pulse and infinite. Pulse irradiation has an irradiation time shorter than the shortest half-life. Infinite irradiation has an irradiation time longer than several times the longest half-life. After irradiation, the delayed neutrons from the sample are counted as a function of time. The delayed neutron count rate as a function of time after irradiation, $\dot{n}_d(t)$, is given as:

$$\dot{n}_d(t) = \epsilon \sum_{i=1}^I P_{n,i} \lambda_i N_i(t), \quad (2.2)$$

where,

$\dot{n}_d(t)$ = the delayed neutron count rate $\left[\frac{n}{s} \right]$

I = the total number of DNPs [-]

ϵ = the detector efficiency [-].

Once this delayed neutron count rate is generated, the data is used to generate group and kinetics parameters for that fissile nuclide and neutron energy. The specifics of the methods used to form those parameters is discussed in Section 2.1.2. Figure 2.1 (recreated from [66]) shows an experimental setup for collecting delayed neutrons.

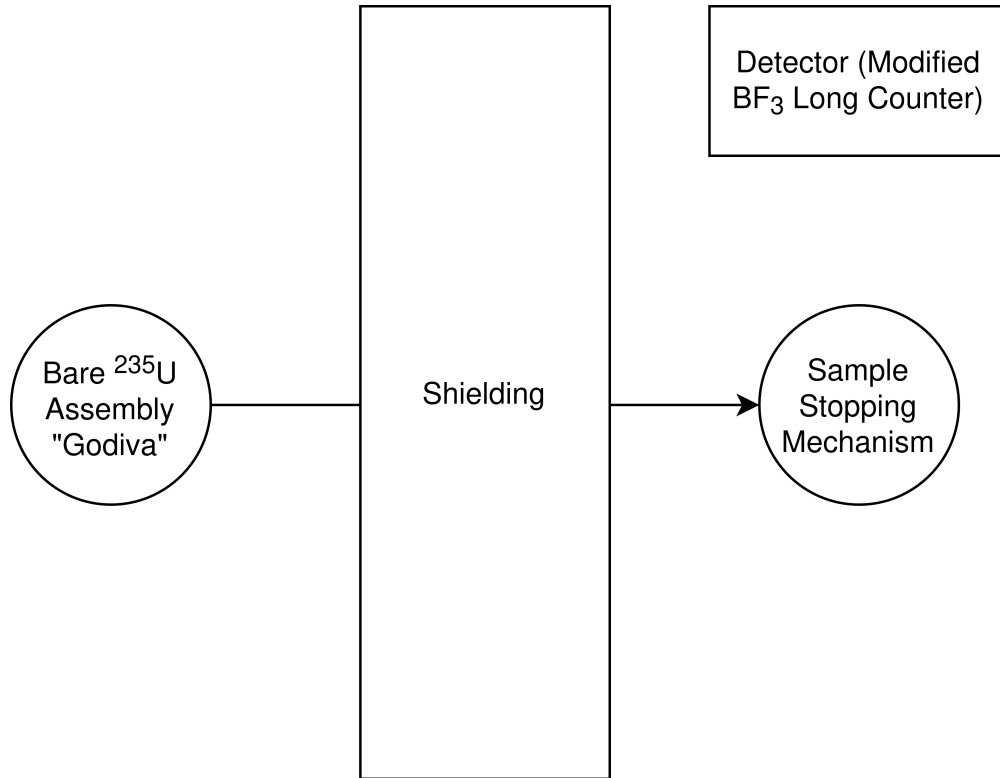


Figure 2.1: Simplified experimental setup for measuring delayed neutron emission from irradiation of a 3 gram sample in Godiva referenced from [66].

Challenges associated collecting this experimental data [10] include experimental noise and signal obfuscation due to moderation. Experimental noise becomes significant after long periods of time as the delayed neutron counts decrease. Additionally, the short lived groups cannot be measured directly due to moderated neutrons from irradiation. Both the long and short time-scale measurements have greater uncertainty than the middle time-scale measurements.

Similarly, uncertainty arises from the measurement of the number of fissions. Although the number of fission events can be measured somewhat accurately using a fission chamber after irradiation or other techniques, the fission chamber does not account for source multiplication within the irradiated sample. Keepin et al. suggests a technique to correct for the sample self-multiplication. The sample self-multiplication require a correction factor of up to 3% in the number of fissions.

Microscopic Measurement of Delayed Neutron Precursors

The other approach used to measure DNPs is the microscopic approach, investigated in detail in the 1990-2000's [106, 32, 123, 146, 20], with progress through the 2010's with more data [92, 62]. Leconte et al. used this approach as recently as 2020 [77]. This approach requires experiments on individual DNP nuclides, from which emission probabilities and half-lives can be measured. Typically, this involves rapid chemical extraction or isotope separation and magnetic tape to rapidly measure short lived nuclides. The magnetic tape is used to transport samples or remove longer-lived decay products [129, 124]. Experimental methods include the neutron- β coincidence technique, separate counting of neutrons and β particles, separate counting of neutrons and γ rays, ion counting, fission yield from a set number of fissions, gamma counting of a long-lived daughter, and normalization of the neutron counter

using a known precursor [124].

There are many different experimental methods used to study DNPs, particularly their spectra: on-line mass spectrometry, helium-3 spectrometry, the proton-recoil technique, the time-of-flight method, and the neutron scintillator [33]. On-line mass spectrometry combined with a helium-3 ionization chamber and the proton-recoil technique allows for measurement of the delayed neutron energy spectrum [128, 16]. The time-of-flight method involves measuring the velocity of the delayed neutron by having a known travel distance and time measurement [29, 47, 28]. An example of the time-of-flight experimental setup is shown in Figure 2.2.

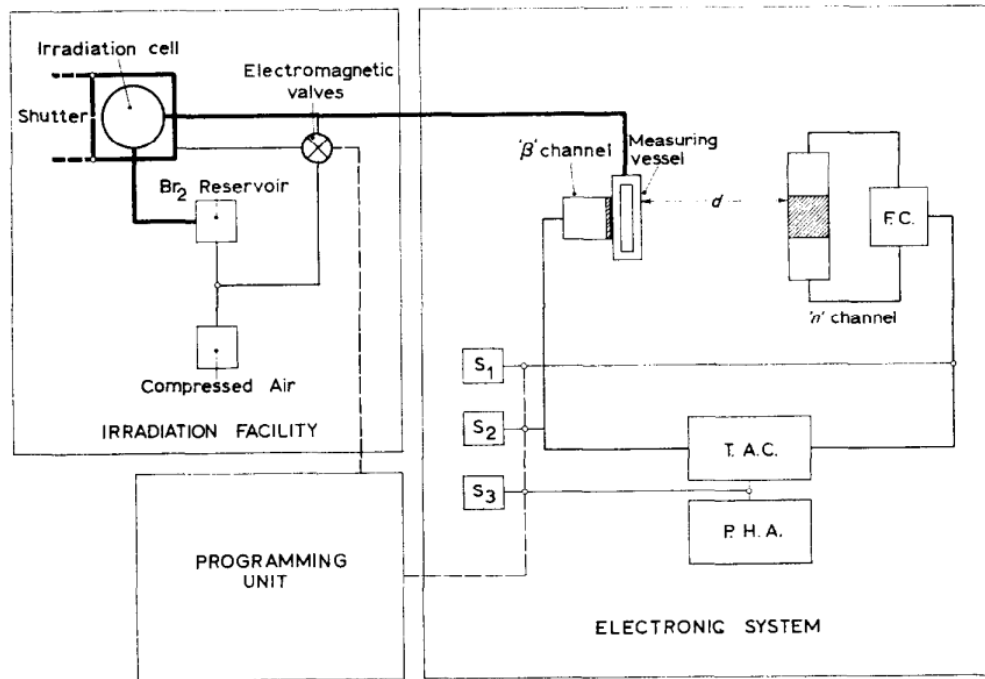


Figure 2.2: A block diagram of the time-of-flight experimental setup for measuring delayed neutrons from bromine-87 and bromine-88, extracted from the Democritos reactor via a fast radiochemical technique recreated from [28].

This microscopic measurement data is exceptionally useful, as it allows for treatment of DNPs on a nuclide-by-nuclide basis. This is particularly important for MSRs, where the spatial DNP concentration depends on half-lives, yields, thermo-chemistry, and thermal-hydraulics. With data on individual nuclides, higher fidelity treatment of DNPs becomes possible.

There have been many new measurements in the past several decades on these nuclides. The IAEA has compiled and evaluated this data in a publicly accessible form [37, 36]. This data, when combined with cumulative fission yields and neutronics data, can be used to model the DNPs explicitly rather than using groups. This is particularly useful for understanding what nuclides have the greatest impact, as well as for simulating experimental configurations which have yet to be thoroughly investigated for DNP group fitting.

2.1.2 DNP Group and Kinetics Parameters

In Section 2.1.1, the various methods for experimentally understanding delayed neutron precursors were discussed. The next stage of understanding these precursors is to understand how they are used in kinetics models, as well as the various ways they are represented. The DNPs are commonly represented in six groups. These groups were developed originally for computational efficiency and due to lacking data for individual precursors. Brady and

England found that increasing the number of groups for point kinetics calculations from six to nine caused no significant differences [20]. Their study shows that increasing the number of DNP groups alone does not provide an increase in accuracy for point kinetics simulations.

This section goes through the calculation of DNP group parameters and kinetics parameters. The DNP group parameters directly affect many of the kinetics parameters, and both sets of data are used in kinetics and dynamics problems. Table 2.1 goes through each of the terms of importance for this section.

Parameter	Description
λ_k/λ_i	The decay constant for DNP group k / nuclide i
$\bar{\nu}_{d,k}/\bar{\nu}_{d,i}$	The delayed neutron yield for DNP group k / nuclide i
χ_k/χ_i	The spectra of DNP group k / nuclide i
$P_{n,i}$	The delayed neutron emission probability for nuclide i
$< T >$	The mean half-life
Y	The total delayed neutron yield
β_k/β_i	The delayed neutron fraction of DNP group k / nuclide i
β	The total delayed neutron fraction
β_{eff}	The effective delayed neutron fraction
Λ	The mean neutron generation time
Λ_{eff}	The effective mean neutron generation time

Table 2.1: DNP group parameters and kinetics parameters along with their definitions.

The DNP groups come with associated parameters, such as the group decay constants, λ_k , and the group yields, $\bar{\nu}_{d,k}$. These group yields can be converted to group delayed neutron fractions, β_k , by dividing the group yields by the average number of neutrons emitted per fission event. The group yields represent the number of delayed neutrons produced per fission in group i , and when summed yield the total delayed neutron yield per fission. The β_k and λ_k terms are the defining parameters for the DNP groups, along with the group spectra, χ_k , for DNP group k . With this data, the other group parameters can be derived, such as the total delayed neutron yield, Y , and delayed neutron fraction β .

Although the groups are useful, there are times when individual DNPs are needed. For example, in 1986, Perry et al. demonstrates that kinetics calculations yield different results when using individual nuclides compared to the six-group parameters [104]. In 1989, Brady and England also use the individual nuclides to compare with group data [20], and updated the six-group parameters based on the explicit DNPs. Updating the six-group parameters enabled greater consistency between experimental results and simulations.

Moving from the group parameters to the kinetics parameters, there is a convenient analogy which can be used to better understand some of the kinetics parameters. Arguably the most well known term in nuclear engineering is the effective multiplication factor, k_{eff} . This term represents the scaling of the number of neutrons over each generation. The infinite multiplication factor, k_{∞} , represents the multiplication without considering leakage terms. This is analogous to the delayed neutron fraction and mean generation time terms, where the effective form of the term requires information on the specific reactor geometry.

The adjoint flux is important for calculating the effective form of many kinetics parameters. The adjoint flux is not a physical quantity, but rather serves as an importance, or weighting, function, describing the importance of the neutron flux in the reactor. For example, areas further from the center of the core are less important than those closer to the center, which is reflected by a decrease in the effective delayed neutron fraction in MSRs as DNPs move

away from the center of the core due to fuel salt advection.

The delayed neutron fraction, β , represents the ratio of delayed neutron production over all neutron production [91]. Related to the delayed neutron fraction is the effective delayed neutron fraction, β_{eff} . This term is more sophisticated, as rather than being a physical property of a fissile nuclide, it depends on the composition and geometry of the reactor. For MSRs, it also accounts for the movement of the DNP in different parts of the reactor.

A straightforward way to interpret the effective delayed neutron fraction is that it is the delayed neutron fraction corrected for the actual composition and geometry of the reactor. More specifically, the effective delayed neutron fraction represents the ratio between the adjoint neutron flux and spectrum-weighted delayed-neutron induced fission to the total number of fission events [120]. Alternatively, it can be thought of as the ratio of the spectrum weighted and adjoint weighted delayed neutron production over all spectrum weighted and adjoint weighted neutron production [91]. For MSRs, the movement of the fuel salt also affects the effective delayed neutron fraction, as the integrated adjoint neutron flux is reduced due to movement of the DNPs out of the core and into less neutronically-important regions.

Another term, the mean neutron generation time, Λ , is the mean time between generation of a neutron and fission induced by that neutron [120]. This term also has an effective form, Λ_{eff} , which is the adjoint weighted form. The effective mean neutron generation time can vary significantly, on the order of hundreds of percent, from the mean neutron generation time [139]. In contrast, the effective delayed neutron fraction varies by on the order of 20% from the delayed neutron fraction [139].

2.1.3 DNP Group Parameters

The DNP group parameters are important for simulating transients. The DNP group parameters are formulated based on the fitting of the delayed neutron count rate, as discussed in Section 2.1.1. Equation (2.2) shows the exact formulation for the delayed neutron count rate as a function of time measured by a detector for I individual DNPs for any number of fissile nuclides, recreated here as:

$$\dot{n}_d(t) = \epsilon \sum_{i=1}^I P_{n,i} \lambda_i N_i(t)$$

Equation (2.3) instead gives the K group DNP fit measured by the same detector for a pulse irradiation as:

$$\dot{n}_d(t) \approx \epsilon F_s \sum_{k=1}^K \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}. \quad (2.3)$$

For an infinite irradiation (also referred to as a saturation irradiation), Equation (2.4) is used rather than Equation (2.3), given as [21]:

$$\dot{n}_d(t) \approx \epsilon \dot{F}_s \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t}. \quad (2.4)$$

The terms in Equations (2.3) and (2.4) are defined as follows:

- K = the number of DNP groups [-]
- λ_k = the decay constant of DNP group k [s^{-1}]
- F_s = the total number of fission events in the pulse irradiation [-]
- $\dot{F}_s$ = the fission rate for a saturation irradiation [s^{-1}]
- $\bar{v}_{d,k}$ = the group yield for group k [-] .

More information on the derivation of these equations can be found in Appendix A. These equations are specifically for a single fissile nuclide, and the derivation can be found in the work of Wilson and England [146].

The pulse irradiation approach gives better statistics on the short lives DNPs, while the longer lived DNPs are better represented by saturation irradiation. Brady discusses this, and uses both pulse and saturation irradiation to form groups [21]. Specifically, Brady uses the four long-period groups from the saturation irradiation and the four short-period groups from the pulse irradiation renormalized at the second DNP group.

Once the delayed neutron count rate is acquired, an over-determined non-linear least squares approach is used to calculate the group yields and decays. For example, Brady used STEPIT to handle the non-linear least-squares solve with both pulse and infinite irradiation experimental data [21]. Waldo et al. also solve the least-squares problem, though the specific approach used is not specified [145]. The general form of the solve is given as:

$$A\vec{x} \approx \vec{b}. \quad (2.5)$$

Equation (2.6) gives the χ^2 term the non-linear least squares minimizes as:

$$\chi^2 = \sum_{m=0}^M \left(\frac{\vec{b}^{(m)} - (A\vec{x})^{(m)}}{\vec{\sigma}^{(m)}} \right)^2, \quad (2.6)$$

where,

χ^2 = the chi-squared measure of the fit

M = the total number of time steps over which the sample decays

σ = the uncertainty at time step m , or unity if the uncertainty is not measured.

Based on the irradiation type used, the coefficient matrix will change, as shown in Equations (2.3) and (2.4). Equation (2.7) gives the coefficient matrix for a pulse irradiation as:

$$A = \begin{bmatrix} \lambda_1 e^{-\lambda_1 t_0} & \lambda_2 e^{-\lambda_2 t_0} & \dots & \lambda_n e^{-\lambda_n t_0} \\ \lambda_1 e^{-\lambda_1 t_1} & \lambda_2 e^{-\lambda_2 t_1} & \dots & \lambda_n e^{-\lambda_n t_1} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1 e^{-\lambda_1 t_m} & \lambda_2 e^{-\lambda_2 t_m} & \dots & \lambda_n e^{-\lambda_n t_m} \end{bmatrix}, \quad (2.7)$$

though for a saturation irradiation the only change would be removing the λ_i term.

Within this least squares equation, the solution vector contains only the group yields, given in Equation (2.8) as:

$$\vec{x} = \begin{bmatrix} \tilde{v}_{d,1} \\ \tilde{v}_{d,2} \\ \vdots \\ \tilde{v}_{d,K} \end{bmatrix}. \quad (2.8)$$

The target vector contains the linearized delayed neutron count rate data normalized to the number of fissions, or fission rate, and the detector efficiency, as shown in Equation (2.9) as:

$$\vec{b} = \begin{bmatrix} \ln(n_d(t_0)(\epsilon F_s)^{-1}) \\ \ln(n_d(t_1)(\epsilon F_s)^{-1}) \\ \vdots \\ \ln(n_d(t_m)(\epsilon F_s)^{-1}) \end{bmatrix}. \quad (2.9)$$

By solving this non-linear system of equations, the group half-lives, T_k , and yields, $\bar{v}_{d,k}$, can be generated. Once this group data is generated, and if there are multiple fissioning nuclides, F , then Equation (2.10) can be used to determine the K group half lives using [33] as:

$$\lambda_k = \frac{\sum_F \lambda_k^{(F)} \beta_{k,eff}^{(F)}}{\sum_F \beta_{k,eff}^{(F)}}, \quad (2.10)$$

where,

$$\begin{aligned} \lambda_k^{(F)} &= \text{the decay constant of DNP group } k \text{ from fission of nuclide } F \\ \beta_{k,eff}^{(F)} &= \text{the effective delayed neutron fraction of DNP group } k \text{ from fission of nuclide } F. \end{aligned}$$

These equations thus far have not included the energy spectra, but instead have focused on the delayed neutron count rate. Equation (2.11) shows that including the energy in the delayed neutron count rate requires the energy spectra, $\chi_{d,k}(E)$, of the relevant group to be included. This is the general form of the saturation and pulse irradiation equations, where A_k represents $\bar{v}_{d,k} \lambda_k$ for a pulse irradiation or $\bar{v}_{d,k}$ for saturation irradiation. $\bar{F}_s$ is the total number of fissions for a pulse irradiation or the fission rate for saturation irradiation. The generic energy and time dependent neutron count rate is given as:

$$\dot{n}_d(E, t) \approx \epsilon \bar{F}_s \sum_{k=1}^K \chi_{d,k}(E) A_k e^{-\lambda_k t}, \quad (2.11)$$

where,

$$\begin{aligned} \bar{F}_s &= \text{the generic fission term [- or } s^{-1} \text{]} \\ A_k &= \text{the generic matrix term [} s^{-1} \text{ or -]} \\ \chi_{d,k}(E) &= \text{the delayed neutron energy spectrum of DNP group } k \text{ [-]}. \end{aligned}$$

2.1.4 DNP Group Spectra and Concentrations

Another consideration in calculating group parameters is constructing group concentrations from individual nuclide data. This is important for situations where reducing the problem from several hundred DNPs to a few DNP groups reduces computational cost. A specific use case for this is explicitly modeling all DNPs spatially and then using those spatially resolved concentrations with an existing six-group structure. In such a case, the DNPs must be combined in such a manner to preserve their contributions to each DNP group. The reverse is also true, where a given six-group spatially resolved concentration can be separated into individual DNPs spatially resolved concentrations.

Two different methods are considered for separating and combining the DNP groups into individual nuclides and vice versa. These methods have been used to generate DNP group spectra, so the group spectra can also be calculated using these methods:

- Grouping based on half lives [40, 121]

- Grouping based on fractional contribution to adjacent groups [21]

The grouping based on half lives requires comparing each nuclide's half life to a constant value. The groupings based on half lives are given in the top two sets in Table 2.2. The bottom set of half lives is from ENDF/B-VIII.0 and is used with the fractional contribution approach [23].

Table 2.2: Half-lives of DNP groups from various sources.

Source	T_1 [s]	T_2 [s]	T_3 [s]	T_4 [s]	T_5 [s]	T_6 [s]
[40]	$T > 31$	$31 \geq T > 9$	$9 \geq T > 3$	$3 \geq T > 0.8$	$0.8 \geq T > 0.3$	$0.3 \geq T$
[121]*	$T > 30$	$30 \geq T > 10$	$10 \geq T > 4$	$4 \geq T > 1.4$	$1.4 \geq T > 0.4$	$0.4 \geq T$
[23]	$T = 51.98$	$T = 21.17$	$T = 5.74$	$T = 2.29$	$T = 0.82$	$T = 0.24$

* Rudstam uses $T_1 = 55.6$, but to ensure all DNPs are included, this has been modified.

The half-life grouping method thus results in a series of comparisons; if a nuclide is within a given group's bounds, the concentration of that nuclide is added to that group.

The fractional grouping is more complex, but assigns nuclides to groups in a more rigorous manner. The method uses two groups, i and $i + 1$, for each nuclide k . These groups are determined via Equation (2.12) as:

$$\lambda_i < \lambda_k < \lambda_{i+1} \quad (2.12)$$

Nuclides which don't fit in these bounds will have the entire fractional contribution go to the closest group. Otherwise, the nuclide will have some fraction of its concentration or spectra go to each DNP group. The fractional contribution, or weight, of nuclide k to each group is represented with $f_{k,i}$ and $f_{k,i+1}$. The total fraction must be equal to 1 as shown in Equation (2.13) as:

$$f_{k,i} + f_{k,i+1} = 1 \quad (2.13)$$

To solve for the fractions, the least squares minimization problem is solved as:

$$\int_0^\infty \left(\lambda_k e^{-\lambda_k t} - f_{k,i} \lambda_i e^{-\lambda_i t} - f_{k,i+1} \lambda_{i+1} e^{-\lambda_{i+1} t} \right)^2 dt \quad (2.14)$$

Solving this equation for where the partial derivatives are zero for each fraction, $f_{k,i}$ and $f_{k,i+1}$, combined with Equation (2.13), yields the overdetermined system of equations as:

$$A = \begin{bmatrix} \lambda_i & \frac{2\lambda_i \lambda_{i+1}}{\lambda_i + \lambda_{i+1}} \\ \frac{2\lambda_i \lambda_{i+1}}{\lambda_i + \lambda_{i+1}} & \lambda_{i+1} \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} \frac{2\lambda_i \lambda_k}{\lambda_i + \lambda_k} \\ \frac{2\lambda_k \lambda_{i+1}}{\lambda_k + \lambda_{i+1}} \\ 1 \end{bmatrix}, \quad x = \begin{bmatrix} f_{k,i} \\ f_{k,i+1} \end{bmatrix}. \quad (2.15)$$

Because the number of rows is greater than the number of columns in the matrix, the left-inverse is used. This system of equations is then solved as shown in Equation (2.16) as:

$$x = (A^T A)^{-1} A^T b \quad (2.16)$$

With the fractional contribution terms, x , the concentration of a given nuclide can then be multiplied by each fraction to show how much it contributes to that group. With a summation of each nuclides contribution, the group concentrations can thus be calculated and vice versa. Alternatively, the spectra can be calculated by using the same approach but with individual DNP spectra instead of concentrations.

2.1.5 Kinetics Parameters

The other relevant parameters to be calculated are simpler and do not require solving least squares problems. The k^{th} DNP group delayed neutron fraction is calculated as:

$$\beta_k = \frac{\bar{v}_{d,k}}{\bar{v}}. \quad (2.17)$$

A useful term for comparing DNP group parameters is the average half-life [107, 108]. Equation (2.18) gives the average half-life as:

$$\langle T \rangle = \ln(2) \sum_k^K \frac{\bar{v}_{d,k}}{\lambda_i}. \quad (2.18)$$

Equation (2.19) shows how the total delayed neutron yield, or average number of delayed neutrons per fission event, $\bar{v}_d$, can be calculated [51]:

$$\bar{v}_d = \sum_k^K \bar{v}_{d,k}. \quad (2.19)$$

This yield can also be approximately represented for an individual nuclide by integrating Equation (2.1) over all time, where the concentration over time is assumed to be:

$$N_i(t) = N_i(0)e^{-\lambda_i t}. \quad (2.20)$$

This yields Equation (2.21), where the total number of delayed neutrons emitted from a single nuclide i is given as:

$$\int_0^\infty \dot{n}_{d,i}(t) dt = P_{n,i} N_i(0). \quad (2.21)$$

The delayed neutron fraction can be written as:

$$\beta = \frac{\bar{v}_d}{\bar{v}} \quad (2.22)$$

The mean neutron generation time can be calculated as:

$$\Lambda = \frac{l}{k}, \quad (2.23)$$

where,

l = the prompt neutron lifetime [s]

k = the neutron multiplication factor [-].

(2.24)

This can be calculated using a time-dependent version of a Monte Carlo code along with point reactor kinetics (PRK) [120].

Finally, there are the effective parameters: the effective delayed neutron fraction, β_{eff} , and the effective mean neutron generation time, Λ_{eff} . β_{eff} is the spectrum and adjoint weighted ratio of the delayed neutron induced fission to the induced fissions from all neutrons [120]. This is given in Equation (2.25) as:

$$\beta_{eff} = \frac{\int \Phi^\dagger \chi_d \beta \bar{\nu} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}{\int \Phi^\dagger \chi \bar{\nu} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}, \quad (2.25)$$

where,

- $\Phi^\dagger$ = the adjoint neutron flux [$\frac{n}{cm^2s}$]
- χ_d = the delayed neutron energy spectrum [-]
- χ = the neutron energy spectrum [-]
- β = the delayed neutron fraction [-]
- $\bar{\nu}$ = the average number of neutrons released per fission [-]
- Σ_f = the macroscopic fission cross section [cm^{-1}]
- Φ = the neutron flux [$\frac{n}{cm^2s}$].

Λ_{eff} can also be calculated explicitly by its definition, which is the "...time between the birth of a neutron and the subsequent absorption that induces fission." This is shown in Equation (2.26) as [139]:

$$\Lambda_{eff} = \frac{\int \Phi_0^\dagger \frac{1}{\bar{\nu}} \Phi_0 dE d\vec{\Omega} d\vec{r}}{\int \Phi_0^\dagger F_0 \Phi_0 dE d\vec{\Omega} d\vec{r}} \quad (2.26)$$

where,

- $\Phi_0^\dagger$ = the critical adjoint neutron flux [$\frac{n}{cm^2s}$]
- Φ_0 = the critical neutron flux [$\frac{n}{cm^2s}$]
- $\bar{\nu}$ = the average neutron speed [$\frac{cm}{s}$]
- F_0 = the total fission operator [cm^{-1}].

There are other methods used to calculate β_{eff} and Λ_{eff} in the literature [139]. These include the direct integration, or adjoint weighted, method [67]; the eigenvalue, or perturbation, method [56]; and the iterated fission probability method [17]. Λ_{eff} can also be calculated using the pulsed method [120]. For β_{eff} , there are also the prompt, Meulekamp, and Spriggs methods which can be used [120, 91]. Aufiero et al. have conducted a comparison of the Meulekamp, iterated fission probability, and prompt methods, which show that the iterated fission probability and prompt methods both yield similar results while the Meulekamp method is 200 pcm higher for the Molten Salt Fast Reactor (MSFR) [8].

2.1.6 Kinetics and Dynamics Methods

There are several different methods which have been used for analyzing transients in MSRs. These methods include, as of 2018, "...fluid-flow-velocity field coupled multi-group, time-resolved, neutron diffusion; ... point reactor kinetics schemes; and quasi-statics; generally all coupled, tightly or loosely, to a thermal-hydraulics solver" [147]. In general, although these methods have varying accuracy and costs, none of them include MSR specific phenomena

when generating the DNP group parameters.

Multi-Group Diffusion

Multi-group diffusion kinetics methods are among the most accurate, and computationally expensive, approaches applied to transient problems, surpassed only by transport methods. Hurwitz and Ehrlich provide a standard formulation of multi-group diffusion written in 1953 [61].

Examples of software that use fluid-flow-velocity field coupled multi-group include GenFOAM and Griffin/Pronghorn coupling using Multiphysics Object-Oriented Simulation Environment (MOOSE) [44, 134]. Other examples can be found in the review by Wooten et al. [70, 45, 69] and elsewhere in the literature [151, 149, 71, 95, 101]. These studies typically use Picard iteration with multi-group neutron diffusion coupled with thermal-hydraulics to accurately capture the multi-physics effects of the reactor. However, this accuracy also comes with a higher computational cost than simpler methods. The general form of the coupling is given as [147]:

$$\begin{aligned} \frac{1}{\bar{v}_g} \frac{\partial \phi_g}{\partial t} = & \nabla \cdot D_g \nabla \phi_g + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} \\ & + (1 - \beta) \chi_{p,g} \sum_{g'=1}^G (\nu \Sigma_f)_{g'} \phi_{g'} \\ & \sum_{i=1}^I \chi_{d,g,i} \lambda_i C_i - \Sigma_{a,g} \phi_g - \sum_{g' \neq g}^G \Sigma_{s,g \rightarrow g'} \phi_{g'}, \end{aligned} \quad (2.27)$$

and,

$$\frac{\partial C_i}{\partial t} = \beta_i \sum_{g=1}^G (\bar{v}_g \Sigma_f)_g \phi_g - \lambda_i C_i - \nabla \cdot (C_i \vec{u}) + \nabla \cdot \frac{\nu_T}{Sc_T} \nabla C_i, \quad (2.28)$$

where,

- $\bar{v}_g$ = the average number of neutrons produced by fission of group g [-]
- ϕ_g = the scalar neutron flux of group g [$\frac{n}{cm^2s}$]
- D_g = the group g diffusion term [cm]
- G = the total number of energy groups [-]
- $\Sigma_{s,g' \rightarrow g}$ = the macroscopic scattering cross section from group g' to group g [cm^{-1}]
- $\chi_{p,g}$ = the number of prompt neutrons born in group g [-]
- $\chi_{d,g,i}$ = the number of delayed neutrons from precursor group i born in group g [-]
- C_i = the concentration of DNPs in group i [$\frac{atoms}{cm^3}$]
- $\vec{u}$ = the velocity profile [$\frac{cm}{s}$]
- $\Sigma_{a,g}$ = the macroscopic absorption cross section for group g [cm^{-1}]
- ν_T = the eddy viscosity [$\frac{cm^2}{s}$]
- Sc_T = the turbulent Schmidt number [-].

Modified Point Kinetics

PRK are a simpler approach to modeling a transient, with the basic form given as:

$$\frac{dn(t)}{dt} = \frac{\rho(t) - \beta_{eff}}{\Lambda} n(t) + \sum_{i=1}^I \lambda_i C_i(t), \quad (2.29)$$

and,

$$\frac{dC_i(t)}{dt} = \frac{\beta_{eff,i}}{\Lambda} n(t) - \lambda_i C_i(t), \quad (2.30)$$

where,

$$\begin{aligned} n &= \text{the relative neutron density [-]} \\ \rho &= \text{the reactivity [-]} \\ \beta_{eff} &= \text{the effective delayed neutron fraction [-].} \end{aligned} \quad (2.31)$$

The derivation of the PRK equations can be found in the work by Henry, with a fluid-fueled derivation in the work by Lapenta et al. [56, 74]. Modified point kinetics (MPK) are used for MSRs, as standard PRK do not properly handle the movement of the fuel salt.

PRK in general struggle to capture spatial effects, such as flow variation, which is important for MSRs. PRK do, however, give a correct prediction for general evolution trends [147]. Wooten et al. provide four different publications which use MPK in MSR analysis [52, 90, 131, 150].

The work by Guo et al. analyzes unprotected loss of heat sink (ULOHs) and unprotected loss of flow (ULOF) transients in the Molten Salt Actinide Recycler and Transmuter reactor (MOSART). The PRK equations are modified into the delayed point reactor kinetics (DPK), which model precursor concentrations in-core and ex-core, with the primary set of equations given as:

$$\frac{dn(t)}{dt} = \frac{\rho - \tilde{\beta}_{eff}}{\Lambda} n(t) + \sum_{i=1}^I \lambda_i C_{i,c}(t), \quad (2.32)$$

$$\begin{aligned} \frac{dC_{i,c}(t)}{dt} &= \frac{\beta_i}{\Lambda} n(t) - \left(\lambda_i + \frac{1}{\tau_c} \right) C_{i,c}(t) \\ &\quad + \frac{1}{\tau_l} \left(\frac{V_l}{V_c} \right) C_{i,l}(t - \tau_l), \end{aligned} \quad (2.33)$$

$$\frac{dC_{i,l}(t)}{dt} = \frac{1}{\tau_c} \left(\frac{V_c}{V_l} \right) C_{i,c}(t) - \left(\lambda_i + \frac{1}{\tau_l} \right) C_{i,l}(t). \quad (2.34)$$

These equations, shown in Equations (2.32) through (2.34) (recreated from [52, 131]), are also used by Shi et al. in the Reactor Excursion and Leak Analysis Program (RELAP). Shi et al. simulate a load demand change, primary flow transient, secondary flow transient, and reactivity transient in the Molten Salt Breeder Reactor (MSBR) [131]. This approach enables modeling of ex-core decay of DNPs, capturing effects at power and for loss of flow, but fails to capture all oscillatory effects [147].

Intermediate terms used in Equations (2.32) through (2.34) are shown in Equations (2.35) through (2.38) as:

$$\tilde{\beta}_{eff} = \beta_{static} - \beta_{loss}, \quad (2.35)$$

$$\beta_{loss} = \frac{\Lambda(t)}{n(t)} \sum_{i=1}^I \lambda_i C_{i,l}, \quad (2.36)$$

$$\tau_c = \frac{M_c}{\dot{M}(t)}, \quad (2.37)$$

$$\tau_l = \frac{M_l}{\dot{M}(t)}. \quad (2.38)$$

The terms in Equations (2.32) through (2.38) are defined as follows:

$$\begin{aligned} V_l &= \text{the volume of the loop [cm}^3\text{]} \\ V_c &= \text{the volume of the core [cm}^3\text{]} \\ \beta_{static} &= \text{the value of } \beta_{eff} \text{ without flow [-]} \\ M_c &= \text{the mass in the core [kg]} \\ M_l &= \text{the mass in the loop [kg]} \\ \dot{M}(t) &= \text{the mass flow rate } [\frac{kg}{s}]. \end{aligned}$$

A similar method is also used by Suzuki and Shimazu for analyzing a reactivity initiated accident in the FUJI-12 reactor [133]. That work uses a simpler form of the equations, without dividing the precursor concentrations into core and loop regions.

The work by Merle-Lucotte et al. uses the "IPK method," which may stand for "Improved Point-Kinetic method," though this is not explicitly stated [90]. This method uses a fixed mesh for neutronics and a mobile mesh for fluid properties. Merle-Lucotte et al. found that the PRK method without any modifications was able to match the IPK method for general trends and end-states, but was unable to capture reactivity oscillations properly. The IPK method was able to capture these oscillations, but assumes uniform fission and instantaneous heat transfer [90].

The work by Zanetti et al. extends the TRACE code, a finite volume code used for LWR transients [150]. The DNP concentration as a function of space and time is resolved upwind in space and with backward Euler for time. Adjoint-weighted quantities discretized with the trapezoidal rule are then passed to PRK equation, and the set of DNP concentration and the PRK power, or neutron density, equations are resolved using LU factorization.

Another approach to modify the PRK is to alter the kinetics parameters based on a phase-space integration weighted with the adjoint flux [75]. Alternatively, effective flow-inclusive group yields can be calculated, as shown by Haubenreich and used in PRK equations [55].

Quasi-Statics

Quasi-static methods seek to maintain the accuracy of multi-group diffusion approaches while also having the cost savings of PRK. The general approach of these methods is to update PRK parameters with occasional multi-group diffusion simulations to capture changes in flux and DNP spatial distributions [147]. The general formulation used for quasi-static methods is very similar among the various publications that use it [38, 134]. A complete derivation of the quasi-static method can be found in the work by Henry, who discusses the shape and amplitude flux functions, and the work by Devooght, who discusses the generalized quasi-static method [56, 35]. There are also other studies which discuss the quasi-static method in detail [12].

Wooten et al. introduce several different publications which have used quasi-statics [147]. One of these, from 2005 by Rineiski et al., covers the SIMMER code, which couples neutronics, thermal-hydraulics, and structure codes [114]. The neutron flux is constrained in the work by a term containing the phase-space integration and the

weighting function used. The neutron flux is treated as:

$$\Phi(\vec{x}, t) = \phi_A(t)\psi(\vec{x}, t), \quad (2.39)$$

where,

$$\begin{aligned} \Phi &= \text{the neutron flux } [\frac{n}{cm^2s}] \\ \phi_A &= \text{the neutron flux amplitude function [n]} \\ \psi(\vec{x}, t) &= \text{the neutron flux shape function } [cm^{-2}s^{-1}]. \end{aligned}$$

Rineiski et al. solve "shape" steps to handle the neutronics and DNP spatial concentrations. The equations used in the "shape" steps are recreated here in Equations (2.40) and (2.41). These equations are solved by time-discretizing Equation (2.40) into a set of steady-state-like equations, which are then treated using the DANTSYS module of SIMMER [5]. These primary equations are given as:

$$\begin{aligned} \frac{1}{\bar{v}} \frac{\partial \psi(\vec{x}, t)}{\partial t} + \frac{1}{\bar{v}\phi_A} \frac{d\phi_A}{dt} \psi(\vec{x}, t) + M(t)\psi(\vec{x}, t) = \\ (1 - \beta)\chi_p F(t)\psi(\vec{x}, t) + \frac{1}{\phi_A} \chi_d \sum_{i=1}^I \lambda_i C_i(\vec{x}, t) + \frac{1}{\phi_A} Q(\vec{x}, t), \end{aligned} \quad (2.40)$$

and,

$$\frac{\partial C_i(\vec{x}, t)}{\partial t} = \phi_A \beta_i F \psi(\vec{x}, t) - \lambda_i C_i(\vec{x}, t) - \nabla(u(\vec{x}, t)C_i(\vec{x}, t)). \quad (2.41)$$

Within these "shape" steps are also sub-steps referred to as "reactivity" steps. At each of these "reactivity" steps, the PRK parameters and flux amplitude are calculated. The equations used in the "reactivity" steps are recreated here as:

$$\frac{d\phi_A}{dt} = \left(\frac{\rho - \beta_{eff}}{\Lambda} \right) \phi_A + q_{eff} + \sum_{i=1}^I \lambda_i (c_{s,i} + c_{m,i}), \quad (2.42)$$

$$\frac{dc_{s,i}}{dt} = \frac{\beta_{eff}}{\Lambda} \phi_A - \lambda_i C_{s,i}. \quad (2.43)$$

$$\rho = \frac{\langle W(\vec{x}), (\chi F(t) - M(t)) \psi(\vec{x}, t) \rangle}{\langle W(\vec{x}), \chi F(t) \psi(\vec{x}, t) \rangle}. \quad (2.44)$$

$$c_{m,i} = \frac{\langle W(\vec{x}), \chi_d C_{m,i} \rangle}{\Lambda \langle W(\vec{x}), \chi F(t) \psi(\vec{x}, t) \rangle}. \quad (2.45)$$

Intermediate terms used in Equations (2.42) through (2.45) are given as:

$$\frac{\partial C_{m,i}(\vec{x}, t)}{\partial t} = -\nabla(u(\vec{x}, t)C_{s,i}(\vec{x}, t)) - \lambda_i C_{m,i}(\vec{x}, t) - \nabla(u(\vec{x}, t)C_{m,i}(\vec{x}, t)), \quad (2.46)$$

and,

$$\frac{\partial C_{s,i}(\vec{x}, t)}{\partial t} = \phi_A(t) \beta_i F(t) \psi(\vec{x}, t) - \lambda_i C_{s,i}(\vec{x}, t). \quad (2.47)$$

The $C_s(\vec{x}, t)$ term is computed at steady state by setting the left-hand side of both Equation (2.40) and (2.41) to zero as well as the fluid velocity to zero at $t=0$. This allows Equation (2.46) to be solved.

The terms in Equations (2.39) through (2.47) are defined as follows:

- $M(t)$ = the neutron disappearance operator [cm^{-1}]
- $F(t)$ = the neutron production operator [cm^{-1}]
- χ_p = the prompt neutron energy spectra [-]
- $C_i(\vec{x}, t)$ = the concentration of DNP group i [$\frac{atoms}{cm^3}$]
- $Q(\vec{x}, t)$ = the external neutron source [$\frac{n}{cm^3s}$]
- $u(\vec{x}, t)$ = the DNP velocity [$\frac{cm}{s}$]
- $C_{s,i}(\vec{x}, t)$ = the "standard" precursor concentration of group i [$\frac{atoms}{cm^3}$]
- $C_{m,i}(\vec{x}, t)$ = the "movable" precursor concentration of group i [$\frac{atoms}{cm^3}$]
- ρ = the reactivity [-]
- $c_{s,i}$ = the PRK "standard" precursor concentration of group i [$atoms$]
- $c_{m,i}$ = the PRK "movable" precursor concentration of group i [$atoms$]
- q_{eff} = the PRK external neutron source [$\frac{n}{s}$]
- Λ = the neutron generation time [s]
- $W(x)$ = the weighting function (the default in SIMMER
is an adjoint flux computed at steady-state conditions) [-].

2.2 Molten Salt Reactor Fuel Depletion

There are many different MSR designs in the literature, as shown in Table 2.3 [27]. These designs feature a circulating salt containing fissile material. In particular, the circulating fuel salt allows for online chemical reprocessing, causes depletion to vary spatially, leads to movement of DNPs, and requires close coupling of thermal-hydraulics to neutronics for realistic modeling.

MSR Name	Source
FUJI series	[46]
Actinides Molten Salt TransmutER (AMSTER)	[140]
MOSART	[63]
Fast Spectrum Molten Salt Reactor (FS-MSR)	[57]
European MSFR	[89]
Small Mobile Molten Salt Reactor (SM-MSR)	[25]
MSBR	[116]
Molten Salt Reactor Experiment (MSRE)	[117]
Accelerator-Driven Neutron Applications (ADNA) Tier-1 reactor	[19]
Single fluid and two fluid denatured molten salt reactors (DMSRs)	[39, 4]
Thorium molten salt reactor (TMSR)	[97, 88]

Table 2.3: A collection of various MSR designs publicly available in the literature.

There are many publications in the literature which have investigated online chemical reprocessing of MSRs, as

shown in Table 2.4 recreated from [126]. Table 2.4 shows which MSR depletion simulations have modeled DNPs in the "DNP" column, showing most models do not include DNP movement in depletion. The table is sorted by the "Reprocessing" column, which signifies the method(s) used in the different MSR models and is also sorted by the publication year in the "Year" column. The table shows there are many MSR concepts, shown in the "Reactor" column, for which depletion was modeled. The study by Valdez et al. investigates two MSR designs, specifically the MSBR and the Molten Salt Demonstration Reactor (MSDR). These models used a variety of spatial refinement levels, shown in the "Model" column, ranging from unit cells to full 3D models.

Table 2.4: Molten salt reactor models analyzed with methods approximating online reprocessing (recreated from [126]).

Publication	Reprocessing	Model	Reactor	DNP	Year
Valdez et al. [65]	Continuous	Unit Cell	Several	No	2020
Zhuang et al. [153]	Continuous	Full 3D	MSFR	No	2020
Xia et al. [148]	Continuous	Full 3D	MSFR	No	2019
Zhou et al. [152]	Continuous	Full 3D	MSBR	Yes	2018
Aufiero et al. [9]	Continuous	Full 3D	MSFR	No	2013
Rodriguez-Vieitez [118]	Continuous	Unit Cell	ADNA	No	2002
Rykhlevskii [125]	Batchwise	Full 3D	MSBR	No	2019
Betzler et al. [15]	Batchwise	Unit Cell	MSBR	No	2017
Ridley and Chvala [113]	Batchwise	Full 3D	Toy	No	2017
Ahmad et al. [4]	Batchwise	Unit Cell	DMSR	No	2015
Park et al. [100]	Batchwise	Full 3D	MSBR	No	2015
Sheu et al. [130]	Batchwise	Full 3D	MOSART	No	2013
Nuttin et al. [96]	Batchwise	Full 3D	MSBR	No	2005
Fiorina et al. [43]	Mixed	2D Slice	MSFR	No	2012
Nagy et al. [94]	Mixed	Unit Cell	Toy	Yes	2008
Merle-Lucotte et al. [88]	Mixed	Full 3D	TMSR	No	2007
Bauman et al. [53]	Mixed	2D Slice	MSBR	Yes	1971

Although these studies incorporate the effects of online chemical reprocessing on depletion, they do not explicitly model the spatial effect on depletion. Instead, a simple approximation is used to handle the in-core and ex-core regions while accounting for the entire primary salt volume. The reason why simulations do not include fully resolved spatial resolution in depletion calculations is due to the computational cost.

2.2.1 Depletion Simulations

Depletion is the process of fission, transmutation, decay, and the changes to material compositions this process causes. Simulating depletion requires specific information, such as decay constants, fission yields, and fission and

transmutation cross-section data. The fission and transmutation cross-section data can come from a transport calculation, while the other data can be read from a data library [79]. Simulating depletion entails numerically solving the Bateman equations; a generalized form of which is given as [79]:

$$\begin{aligned} \frac{dN_j}{dt} = \sum_{i \neq j} [(\gamma_{i \rightarrow j} \sigma_{f,i} \phi + \lambda_{i \rightarrow j} + \sigma_{i \rightarrow j} \phi) N_i] \\ - (\lambda_j + \sigma_j \phi + \lambda_{r,j}) N_j + \dot{N}_{f,j}, \end{aligned} \quad (2.48)$$

where,

- N_j = the atomic density of isotope j $\left[\frac{\text{atoms}}{\text{cm}^3} \right]$
- $\gamma_{i \rightarrow j}$ = the fractional fission product yield of j from the fission of isotope i
- $\sigma_{f,i}$ = the microscopic fission cross-section of isotope i $[cm^2]$
- ϕ = the spectrum-averaged scalar flux in the fuel region $\left[\frac{\text{neutrons}}{\text{cm}^2 \text{s}} \right]$
- $\lambda_{i \rightarrow j}$ = the decay constant of decay $i \rightarrow j$ $[s^{-1}]$
- $\sigma_{i \rightarrow j}$ = the microscopic transmutation cross-section of reaction $i \rightarrow j$ $[cm^2]$
- N_i = the atomic density of isotope i $\left[\frac{\text{atoms}}{\text{cm}^3} \right]$
- λ_j = the decay constant of isotope j $[s^{-1}]$
- $\lambda_{r,j}$ = the removal constant for removal of isotope j $[s^{-1}]$
- σ_j = the microscopic total transmutation cross-section of isotope j $[cm^2]$
- $\dot{N}_{f,j}$ = the feed rate for isotope j $\left[\frac{\text{atoms}}{\text{cm}^3 \cdot \text{s}} \right]$.

The matrix form is given as:

$$\frac{d\vec{N}}{dt} = A\vec{N}, \quad (2.49)$$

where,

- $\vec{N}$ = the vector of material compositions $\left[\frac{\text{atoms}}{\text{cm}^3} \right]$
- A = the depletion matrix $[s^{-1}]$.

The depletion matrix contains the gain and loss terms for each nuclide in N , which vary continuously as a function of time and depend on the concentrations as well. Further documentation on these terms can be found in the OpenMC documentation, an open source Monte Carlo transport code which has built-in depletion solvers [119].

There are several approaches to solving this equation, such as the Transmutation Trajectory Analysis (TTA) method and the Chebyshev Rational Approximation Method (CRAM) [80]. Other codes may use external software to compute depletion, such as REM, an in-house code used by Nuttin et al. [96] to implement 4th order Runge-Kutta. Codes may also use a power series approximation of the matrix exponential, as discussed by Aufiero et al. [9].

Overall, there are many approaches to solving the depletion equation with different computational costs and levels of accuracy. Building a computational tool to handle depletion requires two components: the depletion matrix and a numerical solver.

2.2.2 Stationary and Circulating Fuel Comparison

For stationary fuel, the Bateman equation as shown previously in Equation (2.48) is sufficient to simulate depletion. However, for the purposes of a streamlined comparison with circulating fuel, the Bateman equation can be rewritten as:

$$\frac{dN_j}{dt} = S(t) - \lambda_{net,j} N_j(t), \quad (2.50)$$

where,

$$\begin{aligned} S &= \text{the source term for nuclide } j, \\ &\text{containing fission, decay, and transmutation sources } [\frac{\text{atoms}}{\text{cm}^3 \text{s}}] \\ \lambda_{net,j} &= \text{the loss term for nuclide } j, \\ &\text{containing decay and transmutation losses } [\text{s}^{-1}]. \end{aligned}$$

Over the course of a depletion step, the depletion matrix is assumed to be constant, so the source term varies in time only due to the effects of the changing compositions of other nuclides. The loss term, $\lambda_{net,j}$, does not vary with time in this equation.

This form of the Bateman equation does not have space or energy resolution, which makes it simpler to understand and compare with the circulating fuel version. When solving this equation, the common approach is to have different material groupings, or blocks, in different spatial regions. Each of these material blocks will have different compositions and neutronic properties, such as flux, used for that entire material. This way, the spatial resolution is represented by the material blocks, and the system of ordinary differential equations can be solved rapidly.

Circulating fuel is more complex, as instead of solving a system of ordinary differential equations, a system of partial differential equations must be solved. The simplified static Bateman equation can be modified to include advection and diffusion terms, yielding:

$$\frac{dN_j}{dt} = S(\vec{r}, t) - \lambda_{net,j} N_j(\vec{r}, t) + \nabla \cdot (D \nabla N_j(\vec{r}, t) - \vec{u}(\vec{r}, t) N_j(\vec{r}, t)), \quad (2.51)$$

where,

$$D = \text{the mass diffusivity } [\frac{\text{cm}^2}{\text{s}}].$$

The depletion matrix is assumed to have no spatial resolution in this equation. Because of this assumption, only the concentrations and material blocks provide spatial resolution to the problem. Spatial refinement can be added to this equation based on the specific approach, such as using a spatially resolved neutron flux.

As previously mentioned, the main challenge faced when solving this problem is the computational cost. However, there are many studies which have attempted to develop methods to get around this problem. One common approach, the scaled flux method, is computationally inexpensive [14]. However, this method does not model the in-core and ex-core regions in a physical manner. Instead, it scales the flux by the fraction of fuel salt in-core compared to ex-core. Another approach is to calculate the steady-state concentration, changing the system of ordinary differential equations into spatial rather than temporal relations [127].

Other approaches, such as the method proposed by Tano et al. using Griffin and Pronghorn, try to model the flow more explicitly [135]. This approach is more computationally expensive than the other approaches that only require

solving a system of ordinary differential equations. However, the literature has shown that including advection in depletion directly affects the equilibrium concentration of ^{135}Xe by varying amounts [135, 42]. Tano et al. show that the ^{135}Xe in the MSFR decreases by approximately 80% when moving from no spatial resolution to a quasi-static method. Fei et al. show that the ^{135}Xe in the MSRE increases by approximately 63% when moving from no spatial resolution to 0D fuel flow. Depletion simulations of MSRs need to include spatially resolved depletion to generate accurate results.

However, including fully resolved spatial resolution in depletion has been referred to as computationally "intractable" [14, 135]. Tracking the ~ 2000 nuclides over space and time requires new methods rather than those used for the time dependent system of ordinary differential equations. Methods for solving the system of partial differential equations require accurately capturing short time scale flow effects, making multi-year long depletion simulations exceedingly expensive.

Numerical stability of the problem, when using an explicit-time solving scheme, also requires the Courant-Friedrichs-Lewy (CFL) condition be satisfied as:

$$\lambda_{CFL} \geq \frac{u\Delta t}{\Delta z}, \quad (2.52)$$

where,

Δt = the time step size [s]

Δz = the length between spatial nodes [cm].

This equation limits the time step size based on the flow rate and spatial mesh size. Although using an implicit time solver can work around this condition, the overall computational cost remains high due to the number of nuclides, the time step size, spatial mesh resolution, and the overall amount of time simulated in depletion.

However, for a short depletion simulation, such as a saturation irradiation of a fuel sample, capturing these effects is computationally tractable. Because the depletion lasts only for several minutes instead of several years, the computational cost is orders of magnitude smaller. A high-fidelity, spatially resolved depletion simulation of an MSR sample irradiation experiment is feasible to generate updated DNP group parameters.

2.3 Safety, Security, and Safeguards

2.3.1 Safety

An important part of ensuring safe reactor operation is making sure safety-related transient events do not disrupt safe reactor operations. These transients can be caused by postulated initiating events (PIEs), which are defined in the IAEA Safety Standard Series No. NS-R-1 [1]. Of particular importance are PIEs which potentially lead to design basis accidents (DBAs). Although No. SSR-2/1 has superseded No. NS-R-1, the PIEs and DBAs for MSRs have not changed [2]. Although there are not specific events listed for MSRs, guidance is provided on categories of events are of interest.

These events include equipment failures, such as pumps failing or pipes rupturing; human error, such as improper maintenance or faulty operator actions; external events, such as an earthquake; and combinations of events. For combinations of events, the IAEA specifies that these events should make sense in combination and in the time frame modeled. For example, if a particular event lasts a long time or has a long lasting effect, then that event could possibly overlap with another unrelated event. Alternatively, dependent events could potentially occur

at the same time.

Safety-related transients for MSRs include ULOHS, ULOF, unprotected transient over power (UTOP), and fuel salt over cooling (FSOC) [81, 31]. An ULOHS transient is when the heat exchanger experiences a failure, possibly caused by a pump failure in the intermediate loop, causing a loss of cooling of the reactor. For the MSFR, Brovchenko et al. have shown that slow ULOHS transients that take place over a minute or longer minimize the increase in fuel salt temperature [22]. An ULOF transient is when the flow rate in the primary loop becomes zero, caused by pump failure in the primary loop. An UTOP transient is a reactivity insertion event, which could be caused by various means, such as removing poisons, adding fissile material, or temperature changes. A FSOC transient is when the fuel salt temperature decreases, which could be caused by an increased heat removal rate from the heat exchanger.

The current work is important for safety-related transients, as the MSR specific DNP group parameters will impact the results of these transients. Depending on the properties of the specific reactor, which affect the group parameters, the static DNP group parameter transient simulation may generate overly or insufficiently conservative results.

2.3.2 Security

The goal of security, or physical protection (PP), is to prevent actors from acquiring information, stealing nuclear material, or committing sabotage [110]. Security is enforced through physical protection systems, which detect, delay, and neutralize the adversarial actor before they can complete their objective [6, 48]. There are several different categories which are important to understanding security: the actor type, the actor capabilities, their objectives, and the strategies they use to achieve said objectives. The main consideration for security in this work is protection of the public by preventing: theft of hazardous radioactive material and sabotage with the intent to damage facilities, disrupt operations, or release radioactive material. The majority of the information presented in this section comes from The Proliferation Resistance and Physical Protection Evaluation Methodology Working Group of the Generation IV International Forum [110].

The actor type involved in PP can be an individual, a group, a large organization, or a non-host state combined with any of the former. These actors may be composed of some combination of insiders and outsiders. Insiders are employees or contract staff who may be sympathetic with or suborned by the hostile actor. Outsiders can include terrorist groups, agents of proliferant states, or lone individuals. Based on the specifics of the actor type, their capabilities will vary significantly.

There are five categories of actor capabilities: knowledge, skills, weapons and tools, number of actors, and dedication. The knowledge of the actor will vary depending on if they have insider knowledge or not, and how much information that insider knows. The skills of the actor will affect their strategy they use to achieve their goals, though a large organization or non-host state can be assumed to have access to a wide variety of skills. The weapons and tools of the actor will affect their strategy, but access to an insider would grant more tools accessed with less effort. The number of actors will vary with the specific strategy of that actor as well as their tactics, as a more discrete approach would require fewer actors than a forceful approach. The dedication of the actor is important to consider, as an insufficiently motivated actor may be deterred with sufficient security measures.

The objectives of the actor may include: disruption of operations, radiological release, nuclear explosives, Radiological Dispersal Devices (RDDs), and/or information theft. The objectives of the actor will influence their other decisions, particularly their strategies. If an actor's objective requires force, then the capabilities and strategies implemented by that actor will vary from one whose objective can be accomplished covertly. Assuming the actor is rational, they will select a strategy which optimizes the probability of successfully completing their objective.

The actor's strategies include their modes of attack and tactics. The modes of attack vary significantly, including

cyber attacks, waterborne attacks, direct or diversionary attack using light anti-tank weapons, and more. The tactics actors can use include stealth, deceit, and overt force. The stealth and deceit tactics are more covert, including avoiding or deactivating security systems for the stealth tactic or using false identification or authorization for the deceit tactic. The overt force tactic ties more directly into many of the modes of attack, which may itself also serve as a diversion for a different tactic.

For this work, the specific actor capabilities and strategies are not considered. Additionally, only a single type of actor and objective combination is considered – an actor with access to an insider and the objective to disrupt operations through sabotage. This work investigates the formulation of new DNP group parameters, which impact the reactor behavior during a transient. There may be security issues that arise during a sabotage-initiated transient due to the recalculated kinetics and DNP group parameters.

2.3.3 Safeguards

Preventing the spread of nuclear material for weapons is the goal of nuclear safeguards. Safeguards is defined as the nuclear reactor characteristics that impede "...the diversion or undeclared production of nuclear material and the misuse of technology by the Host State seeking to acquire nuclear weapons or other nuclear explosive devices." [110]. This definition agrees with the definition made at the IAEA international workshop in 2002 [64]. In this work, proliferation resistance (PR) is used as a way to understand and measure certain criteria used in safeguards. PR is a spectrum, so it should not be considered necessary to optimize the PR, but rather as a way to evaluate strengths and weaknesses in the design relating to safeguards.

There are six measures for PR: technical difficulty, cost, time, fissile material type, detection probability, and detection resource efficiency. In this work, the measures analyzed include time and detection probability. The time measure is analyzed by conducting diversion scenarios over various time frames to determine the effects on diversion fingerprints. A very low PR time is 0-3 months, while a very high PR time is >30 years [110]. The detection probability measure goes hand-in-hand with time, as a protracted diversion scenario is likely to have a reduced detection probability. A very low PR detection probability is 0-5 percent, while a very high PR detection probability is 95-100 percent [110].

The Department of Energy (DOE) released an updated nuclear material control and accountability (MCA) Order in February of 2023 that cancels DOE Order 474.2 Chg 4, which was last updated in 2016 [58]. Based on the more recent Order, special nuclear material (SNM) is classified as shown in Table 2.5.

Table 2.5: The SNM classification table is recreated from [58].

Material Type	Accountable Quantity	Weight Field Used for Element	Weight Field Used for Isotope
^{235}U	1 g	total uranium	^{235}U
^{233}U	1 g	total uranium	^{233}U
^{242}Pu	1 g	total plutonium	^{242}Pu
$^{239-241}\text{Pu}$	1 g	total plutonium	$^{239}\text{Pu} + ^{241}\text{Pu}$
^{238}Pu	0.1 g	total plutonium	^{238}Pu
$^{241}\text{Am}^{**}$	1 g	total americium	^{241}Am
$^{243}\text{Am}^{**}$	1 g	total americium	^{243}Am
$^{237}\text{Np}^{**}$	1 g	total neptunium	-

^{**} Americium and neptunium are treated as SNM if they are separated.

Table 2.5 is used along with the graded safeguards table from the updated Order to properly classify the attractiveness level and categorization of SNM. These categories include weapons, pure products, high-grade materials, low-grade materials, and all other materials, ranging from attractiveness level "A" to "E". For this work, only materials of attractiveness level "C" and below are relevant. High-grade materials, with attractiveness level "C", include fuel elements and assemblies. Low-grade materials, with attractiveness level "D", include solutions with target concentrations of 1-25 g/L. All other materials, with attractiveness level "E", are made up of highly irradiated forms, solutions with concentrations <1 g/L, or uranium containing <20% ^{235}U or <10% ^{233}U .

The quantities provided in Table 2.5 show the minimum reportable, or accountable, quantities for each material type. These quantities make up the lower limit of Category IV materials for "B" through "E" attractiveness level materials. The SNM Category depends on the quantity of material as well as the attractiveness level of that material. Higher Category material has more stringent requirements due to their greater strategic importance, with Category I SNM having the most requirements. For example, if material is classified as attractiveness level "C", then any quantity of plutonium or $^{233}\text{U} \geq 6$ kg is considered Category I SNM. Alternatively, if the same material contains ^{235}U , separated ^{237}Np , or ^{241}Am and ^{243}Am , then an amount ≥ 20 kg is considered Category I SNM.

If the material in the reactor is classified as attractiveness level "D", then any quantity ≥ 16 kg is considered Category II, as Category I quantities do not exist for this attractiveness level. Material of attractiveness level "D" do not have a listed Category I quantity for material containing ^{235}U , separated ^{237}Np , or ^{241}Am and ^{243}Am .

Material of attractiveness level "E" does not have Category I, II, or III quantities. Instead, reportable quantities are considered Category IV.

Significant quantities (SQs) of material are also of interest, of which there are two categories: direct use and indirect use. Direct use nuclear material consists of ^{233}U , highly enriched uranium (HEU) (^{235}U enriched over 20%), and any amount of ^{239}Pu with less than 80% ^{238}Pu . Indirect use nuclear material consists of thorium and low enriched uranium (LEU) (^{235}U enriched at less than 20%). SQs of the different material categories are given in Table 2.6 [110, 3, 11].

This information is important for the current work, as it provides values for quantities of diverted material to analyze. The exact classification, materials, and quantities to use will vary based on the specifics of the fuel salt and diversion scenario. However, this information provides a set of data which can be directly referenced to determine

Use	Material	Significant Quantity [kg]
Direct	Plutonium	8
Direct	HEU	25
Direct	²³³ U	8
Indirect	²³⁵ U	75
Indirect	Natural Uranium	10,000
Indirect	Depleted Uranium	20,000

Table 2.6: Significant quantities of various materials are given along with their use category.

the diversion rates.

For example, a fuel salt categorized as direct use with a plutonium diversion would have 8kg of plutonium as a significant quantity. If that fuel salt also had an attractiveness level of "D", then 16kg of plutonium would be Category II. Then, the diversion time could vary over a range of months to years, based on various PR times. A method of detection for these diversion scenarios is via change in reactivity due to the change in delayed neutron precursors and fissile concentration. The detection probability is then estimated based on this measurable reactivity change.

Chapter 3

Methodology

3.1 Tools

Several external software tools evaluate and generate the updated DNP group parameters. In-house Python scripts collect and combine data, OpenMC generates the group parameters, and the MOOSE app Moltres evaluates and refines them at different fidelity levels [138, 119, 83]. Each tool is described in detail in its corresponding sub-section.

3.1.1 Python Scripts

The Python scripts used in the simplified DNP group generation rely on the SciPy, NumPy, Matplotlib, *pandas*, and Uncertainties packages [142, 54, 60, 136, 41]. These scripts automate the OpenMC sample irradiation simulation, collect data from the IAEA beta-delayed neutron emission database, combine OpenMC concentrations with half-life and emission probability data, and fit group parameters to the calculated delayed neutron count curve [37]. Section 3.2 provides further details on these scripts.

3.1.2 OpenMC

OpenMC is an open-source Monte Carlo neutron and photon transport code. Its Python Application Programming Interface (API) offers functionality that simplifies interfacing with in-house Python scripts. The sample irradiation process is straightforward and configurable, as users can adjust parameters through script inputs. OpenMC also offers useful feature in depletion, where a zero-power depletion step is rapidly calculated since only depletion is tracked. This feature allows resolving ex-core residence time depletion steps quickly. Additionally, OpenMC's transfer rate functionality during depletion enables simulation of chemical removal in an MSR.

For this work, the scripts use the Evaluated Nuclear Data File (ENDF)/B-VIII.0 based depletion chains provided by OpenMC and ENDF/B-VIII.0 cross section data. The thermal spectrum chain applies to neutron irradiation below 300 eV, while the fast spectrum chain applies to higher-energy neutron irradiation. Monoenergetic neutrons irradiate the fissile sample in the 0D flow model. This approach ensures the irradiation generates group parameters for a specific neutron energy rather than a particular spectrum.

3.1.3 Moltres

Overview

MOOSE is an open-source, parallel, finite element framework for multiphysics applications [50]. MOOSE includes many pre-built applications that couple physical phenomena. The HEAT CONDUCTION and NAVIER-STOKES modules are particularly useful in this work, as they can couple with Moltres for general thermal-hydraulic modeling [103, 82, 105, 102].

Moltres is an open-source MOOSE application that simulates MSRs by solving the multi-group neutron diffusion and thermal-hydraulics equations [83]. This work uses Moltres to simulate the MSRE with different DNP group parameters. Moltres also validates the updated DNP group parameters in this work.

The MSRE experimental data can be used for validation, and was previously modeled with Moltres [83]. The MSRE core consists of flow through channels, simplifying its modeling compared to open-pool designs due to the restricted flow path. Since the initial demonstration, Park et al. have noted general agreement in results, attributing differences to simplifications and assumptions in the geometry [102]. Additionally, Park has published a dissertation detailing 1-D, 2-D, and 3-D neutronic models of the MSRE [101].

Moltres can simulate transients using the MSR-specific DNP group parameters, and it can generate updated parameters with higher fidelity than the 0D method. Moltres couples three physical phenomena to model an MSR: neutron flux, DNPs, and thermal-hydraulics.

Flux Modeling

Neutron flux modeling determines the energy of neutrons interacting with the sample and the number of interactions. It determines the fission rate and the resulting fission product concentrations, including DNPs. The time-dependent multi-group neutron diffusion equation appears in Equation (2.27), recreated here:

$$\begin{aligned} \frac{1}{v_g} \frac{\partial \phi_g}{\partial t} = & \nabla \cdot D_g \nabla \phi_g + \sum_{g' \neq g}^G \Sigma_{s,g' \rightarrow g} \phi_{g'} \\ & + (1 - \beta) \chi_{p,g} \sum_{g'=1}^G (v \Sigma_f)_{g'} \phi_{g'} \\ & - \sum_{i=1}^I \chi_{d,g,i} \lambda_i C_i - \Sigma_{a,g} \phi_g - \sum_{g' \neq g}^G \Sigma_{s,g \rightarrow g'} \phi_g \end{aligned}$$

Modeling the neutron flux requires accounting for the DNPs. DNPs contribute to the neutron flux and possess their own energy spectra, necessitating their inclusion as a separate term.

Precursor Modeling

The time dependent DNP concentration is given in Equation (2.28) and recreated here:

$$\frac{\partial C_i}{\partial t} = \beta_i \sum_{g=1}^G (v \Sigma_f)_g \phi_g - \lambda_i C_i - \nabla \cdot (C_i \vec{u}) + \nabla \cdot \frac{v_T}{Sc_T} \nabla C_i$$

An issue arises with this equation because Moltres generates MSR-specific DNP group parameters, yet the equation requires predefined DNP group parameters. To generate the DNP group parameters, this equation is replaced by (2.51), recreated here:

$$\frac{dN_j}{dt} = S(\vec{r}, t) - \lambda_{net,j} N_j(\vec{r}, t) + \nabla \cdot (D_{mix} \nabla N_j(\vec{r}, t) - \vec{u}(\vec{r}, t) N_j(\vec{r}, t))$$

Specifically, the solution to this problem is to use spatially resolved depletion over the irradiation simulation; instead of tracking concentrations of DNP groups, the simulation tracks concentrations of the individual DNPs.

Thermal-Hydraulic Modeling

Moltres offers time-dependent, incompressible, buoyancy-driven flow using Equations (3.1) and (3.2) recreated from [101]:

$$\rho \frac{\partial \vec{u}}{\partial t} = -\rho(\vec{u} \cdot \nabla) \vec{u} - \nabla p + \mu \nabla^2 \vec{u} + \rho \alpha \vec{g} (T - T_{ref}), \quad (3.1)$$

$$\nabla \cdot \vec{u} = 0, \quad (3.2)$$

where,

ρ = fluid density

p = pressure

μ = dynamic viscosity

α = coefficient of thermal expansion

$\vec{g}$ = gravitational force vector

T = fluid temperature

T_{ref} = reference temperature at which the nominal density is provided.

The temperature distribution is given by Equation (3.3) recreated from [101], which shows the temperature advection-diffusion as:

$$\rho c_p \frac{\partial T}{\partial t} = -\rho c_p \vec{u} \cdot \nabla T + \nabla \cdot (k \nabla T) + Q_f - Q_s, \quad (3.3)$$

and by Equation (3.4) recreated from [101], which gives the fission heat source term as:

$$Q_f = \sum_{g=1}^G \epsilon_g \Sigma_g^f \phi_g, \quad (3.4)$$

where,

c_p = specific heat capacity of molten salt

k = effective thermal conductivity of molten salt

Q_f = fission heat source

ϵ_g = average fission energy released by neutrons in group g ,

Q_s = heat sink/removal.

With the combination of the flux modeling, precursor modeling, and thermal-hydraulic modeling, Moltres can almost simulate MSRs. However, there is one additional MSR phenomena to complete the model. This phenomena is the flow of the fuel salt between the in-core to ex-core regions.

Ex-core and In-core Coupling

Moltres treats the ex-core region as a tightly-coupled 1D pipe geometry with coupling using Picard iterations and the inlet/outlet boundary conditions. The specific equations to treat the variables of interest in this geometry can be derived from their full forms. For example, the flux and eddy viscosity are zero in the ex-core region, leading to Equation (3.5), which describes the precursor concentration in the ex-core region as:

$$\frac{\partial C_i}{\partial t} = -\lambda_i C_i - u_x \frac{\partial C_i}{\partial x}. \quad (3.5)$$

3.2 DNP Group Generation

Generating DNP group parameter data in a reactor-agnostic manner requires a reactor-agnostic generation process. Fully detailing the flow regime with a large number of parameters is computationally infeasible. Instead, the flow is simplified to two parameters: an in-core average residence time, τ_{in} , and an ex-core residence time, τ_{ex} .

This approach captures fuel salt advection around the primary loop and approximates turbulent and diffusive mixing. The terms derive from the averaged behavior of a fuel salt control volume, treating thermal-hydraulic effects, such as vortices, as changes in residence times. This method allows for approximate fitting of any circulating fuel reactor, requiring only the generation of relevant parameters for determining the associated DNP group parameters.

The other required parameters describe chemical removal, which varies by fission product and reprocessing rate. Refueling is not considered in this work, as its effects are negligible over the short time frames analyzed. An irradiation time of approximately seven minutes suffices to reach equilibrium DNPs concentrations, as the saturation time frame is on the order of multiple half-lives of the longest-lived DNP. Elements with cycle times on the order of months to years exhibit negligible changes, as discussed in Section 4.1.6. Elements with changes on the order of seconds to days contribute sufficiently and must be included for accurate data generation.

The MSBR reprocessing scheme serves as the basis for cycle times in this work, as shown in Table 3.1. Rare earths, semi-noble metals, volatile fluorides, discard, and higher nuclides have negligible effects over the time scales studied. Section 4.1.6 examines these longer cycle times to confirm their negligible impact. Section 4.3 scales the cycle times to analyze how reprocessing rates influence the group parameters.

Table 3.1: MSBR Online Reprocessing Cycle Times from [116].

Reprocessing Group	Element(s)	Cycle Time (Full Power)
Rare Earths	Y, La, Ce, Pr, Nd, Pm, Sm, Gd	50 days
Rare Earths	Eu	500 days
Noble Metals	Se, Nb, Mo, Tc, Ru, Rh, Pd, Ag, Sb, Te	20 seconds
Seminoble Metals	Zr, Cd, In, Sn	200 days
Gases	Kr, Xe	20 seconds
Volatile Fluorides	Br, I	60 days
Discard	Rb, Sr, Cs, Ba	3435 days
Salt Discard	Th, Li, Be, F	3435 days
Protactinium	Pa	3 days
Higher Nuclides	Np, Pu	16 years

The parameters used include incident neutron energy, target fissile nuclide, in-core residence time, ex-core residence time, reprocessing scheme, and reprocessing scaling vector. Each unique parameter adds a dimension to the data, exponentially increasing the computational cost of generating group parameters for every combination. This work uses the MSBR reprocessing scheme with a single scaling value applied to each reprocessing group instead of a unique scaling value for each. The incident neutron energies that irradiate the sample are 0.0253 *eV* and 1 *MeV*. Section 4.3 determines the minimum number of residence times and their values required to capture general trends.

3.2.1 0D Flow Model

A simplified model is necessary to generate sufficient data for each combination of input parameters. This model estimates the magnitude of variation and identifies general trends. It treats the flow in a zero-dimensional manner using in-core and ex-core residence times. The fission history function is a square wave, with a constant fission rate during the in-core residence time and no fission during the ex-core residence time. This approximation simplifies the analytical form of the fission history for creating the non-linear least-squares problem, as shown in Appendix A.

Figure 3.1 shows how OpenMC uses this fission history to simulate irradiation on a control volume of fuel salt circulating in the primary loop. The model passes eight parameters into OpenMC [119]: in-core residence time, ex-core residence time, neutron irradiation energy, target fissile nuclide, target density, total irradiation time, temperature, and reprocessing terms. Three parameters, shown in green, remain constant: temperature, target density, and total irradiation time. The remaining inputs, shown in yellow, vary to generate new group parameters. Section 3.3 provides further details on this process.

After OpenMC completes the fixed-source simulation, the final nuclide concentrations in the irradiated sample are saved for analysis in custom scripts. These scripts combine the concentrations with data from the IAEA beta-delayed neutron emission database to generate a delayed neutron count rate [37]. A non-linear least squares fitting function from SciPy then calculates the group half-lives and yields [143]. These group parameters correspond

to a specific nuclide, neutron energy, in-core residence time, ex-core residence time, reprocessing scheme, and reprocessing scalar. Section 3.4 provides further details on this block. Some additional factors to consider include self-shielding due to sample density (while avoiding self-multiplication), temperature-induced cross-section variations, and under- or over-irradiation, all of which can affect the results [66].

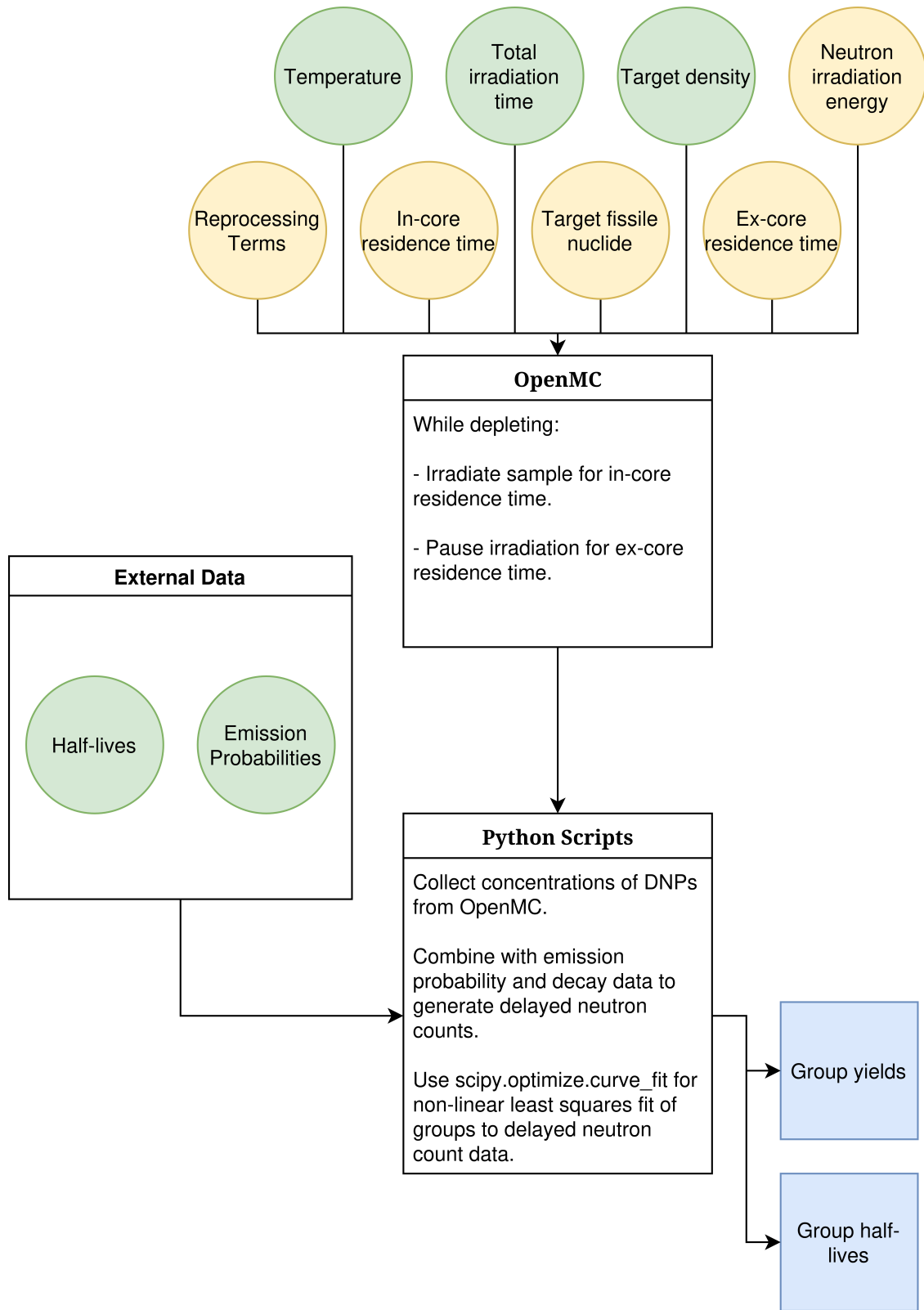


Figure 3.1: Flow diagram of 0D flow model used to generate flow-inclusive precursor group fit parameters.

The OpenMC model uses a three-gram spherical sample with vacuum boundary conditions. It treats the

problem as a fixed-source simulation, with a uniform neutron source distributed evenly throughout the sphere. This approach is not physical, as a real neutron source would have a spectrum and originate externally. However, the exact neutron spectrum for irradiating samples to generate DNP group parameters is not provided in the literature. The exact neutron energy spectrum is not needed, as only significant differences in the neutron energy spectrum affect the DNP group parameters. The source distribution also has negligible impact, as the sample is small and exhibits minimal self-multiplication. Thus, this approximation does not limit the effectiveness of the 0D flow model.

Another consideration for the OpenMC model is maintaining a consistent source particle rate. While not strictly necessary for group parameter generation since the values are per-fission, it simplifies the data analysis, as the tallies depend on the source particles. Equation (3.6) provides the source rate for the irradiation simulations:

$$\dot{S} = S_{net} \frac{\tau_{in} + \tau_{ex}}{\tau_{in} T}, \quad (3.6)$$

where,

- $\dot{S}$ = the source particle rate [$\frac{n}{s}$]
- S_{net} = the total number of source particles [n]
- τ_{in} = the in-core residence time [s]
- τ_{ex} = the ex-core residence time [s]
- T = the net irradiation time [s].

3.2.2 Spatially Resolved Model

The 0D flow model rapidly generates DNP group parameters, but it simplifies the treatment of flow. Instead of modeling the actual fuel salt flow, it estimates fuel circulation using residence times for each region. Figure 3.2 shows the flow diagram for a spatially resolved model using Moltres instead of OpenMC.

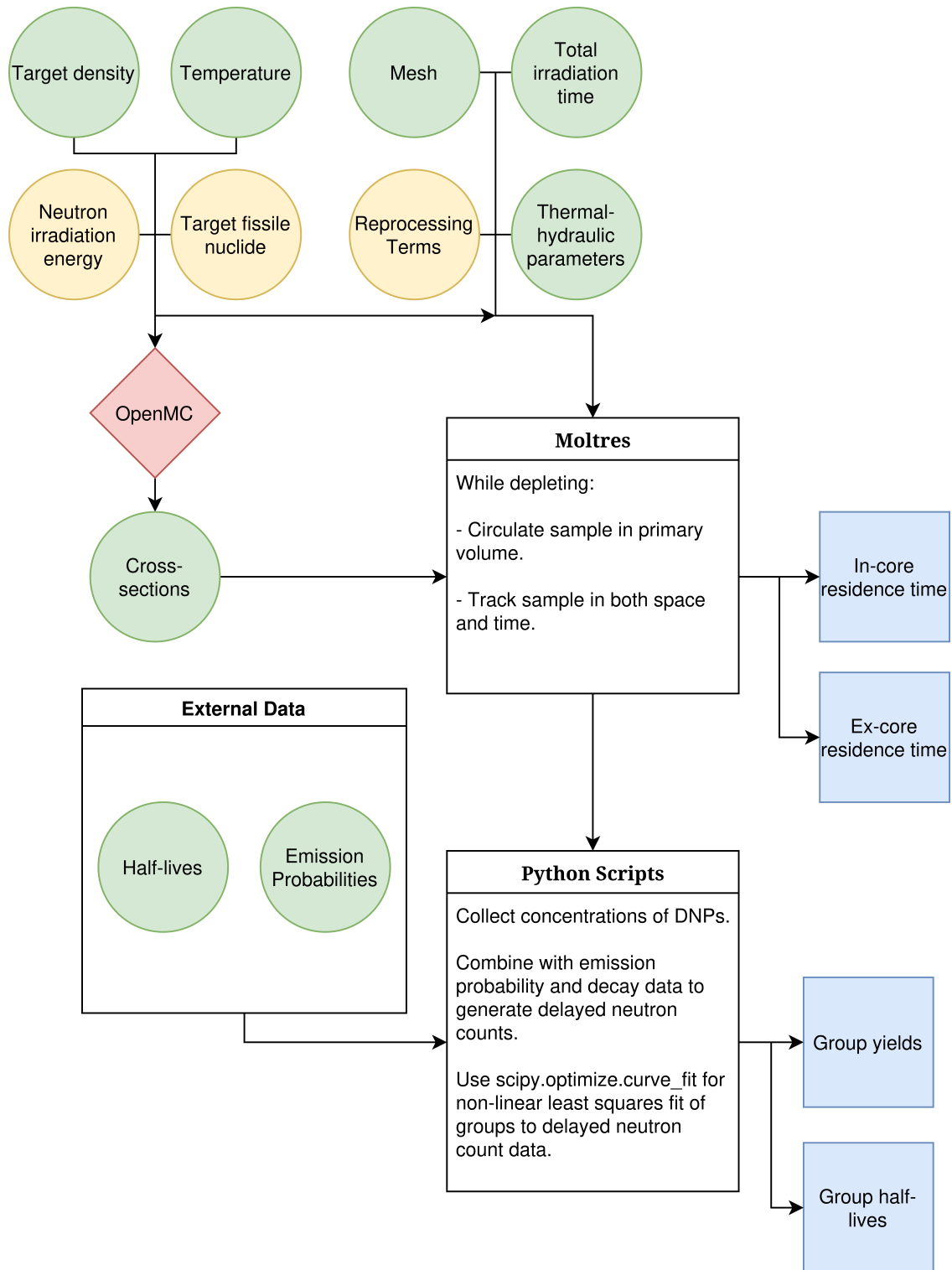


Figure 3.2: Flow diagram of spatially resolved flow model used to generate flow-inclusive precursor group fit parameters.

The spatially resolved flow model introduces two new parameters absent in the 0D flow model. The first is the mesh, which defines the problem geometry. MOOSE provides built-in meshing tools, though complex geometries

may require external tools such as Gmsh [49]. The second is the thermal-hydraulics parameters, which capture physical phenomena enabled by spatial resolution, as discussed in Section 3.1.3.

In the spatially resolved model, the in-core and ex-core residence times are outputs rather than inputs. These times derive from the movement of the irradiated sample over time in the fuel salt. The in-core and ex-core residence times vary as the irradiated sample diffuses and mixes further into the fuel salt. The net residence time, τ_{net} , remains constant unless the model changes, such as an increased temperature difference or induced flow rate. Equation (3.7) defines the in-core residence time as:

$$\tau_{in}(t) = \frac{\int_{V_{in}} \int_0^t N(\vec{x}, t) dt d\vec{x}}{\int_V \int_0^t N(\vec{x}, t) dt d\vec{x}} \tau_{net}, \quad (3.7)$$

where

$$\begin{aligned} V_{in} &= \text{the in-core volume [cm}^3\text{]} \\ V &= \text{the total primary volume [cm}^3\text{]} \\ N(\vec{x}, t) &= \text{the time and space resolved sample concentration [} \frac{\text{atoms}}{\text{cm}^3} \text{]} \\ \tau_{net} &= \text{the net residence, or circulation, time of the fuel salt [s].} \end{aligned}$$

The ex-core residence time can be calculated in the same way, or alternatively as shown in Equation (3.8) as:

$$\tau_{ex} = \tau_{net} - \tau_{in}. \quad (3.8)$$

3.3 Delayed Neutron Count Rate Generation

Although the 0D flow model overview has been discussed, execution details were not. The following sub-sections provide specific information on the methodology behind the OpenMC block. This includes a detailed discussion on the concentrations generated by OpenMC and the data used with those concentrations to produce a delayed neutron count rate.

3.3.1 DNP Concentrations

The OpenMC model uses a fixed-source problem on a three-gram spherical sample of a fissile target. A uniformly distributed source is placed inside the target with a vacuum boundary condition at the edge of the sphere. The sample must have a sufficiently large volume to ensure a high number of fissions, but excessive volume increases self-shielding and self-multiplication. Adjusting the density mitigates this issue: higher density increases self-shielding and self-multiplication, while lower density reduces fission probability and worsens statistical accuracy.

The fixed-source in this model consists entirely of neutrons at a specific energy, rather than a spectrum. This energy dictates fission cross sections, transmutation cross sections, and fission yields. Traditionally, tabulated group parameter data is formatted for a single energy or energy range, such as fast or thermal. However, macroscopic experiments, discussed in Section 2.1.1, use reactors with varying neutron energy spectra, not mono-energetic neutrons. This could be addressed by modeling a reactor in OpenMC and using a k-eigenvalue solve to generate a neutron energy spectrum. However, this work assumes the spectral effects are sufficiently small to exclude. Photons can also generate delayed neutrons through photofission, though this is not investigated here [68].

The neutron source distribution does not replicate the experimental process, where neutrons irradiate the

sample externally rather than internally. However, an evenly distributed source eliminates self-shielding concerns. The exact spatial locations of fission events within the sample should negligibly affect the nuclide concentrations, which are the outputs of the OpenMC simulation. With a density of $10 \frac{g}{cc}$, the spherical sample has a radius of approximately 0.4 cm . The spatial resolution of concentrations within the sample is insignificant. Additionally, the concentrations extracted from the OpenMC simulation are spatially collapsed in the post-processing, so the spatial effect of a uniformly distributed source serves only to reduce any self-shielding effects which could reduce the statistics and cause more transmutation than expected.

After running the OpenMC simulation, the concentrations are collected from the depletion results for the final irradiation time step. These final concentrations are passed to other scripts that convert the concentrations into a delayed neutron count rate.

3.3.2 Concentrations to Count Rate

After receiving the concentrations from OpenMC, the scripts determine the concentration of each nuclide as a function of time during the measurement period. The concentrations are calculated based on half-life, as shown in Equation (3.9), where the concentration as a function of time is:

$$N(t) = N_0 e^{-\lambda t}. \quad (3.9)$$

This simple equation does not account for decay chains between nuclides. An alternative approach uses repeated zero-power depletion simulations in OpenMC to generate concentrations as a function of time, incorporating decay chains. These concentrations are available at coarse time steps and linearly interpolated using SciPy, allowing the scripts to access more accurate concentrations at any time after irradiation [141]. This approach only models delayed neutrons generated during decay, as it is beyond the scope of this work to model other particles [26]. Section 4.1.7 compares these two approaches.

The concentrations are combined with Equation (2.2), recreated here:

$$\dot{n}_d(t) = \epsilon \sum_{k=1}^K P_{n,k} \lambda_k N_k(t).$$

The efficiency term, ϵ , is set to 1 in this work, as the simulated method includes every produced delayed neutron in the count rate. The contributions of each DNP over time are summed to generate the delayed neutron count rate.

3.4 DNP Group Reconstruction

Once the delayed neutron count rate is generated, it is converted into DNP group parameters. Fitting the group parameters requires a non-linear least squares method, as both the group half-lives and yields must be determined. The equation describing the problem appears in Equation (2.3), recreated here:

$$\dot{n}_d(t) \approx \epsilon F_s \sum_{k=1}^K \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}.$$

An important consideration for this process is the selection of time values for generating the group fits. Increasing the number of time nodes raises computational costs, but too few nodes result in a poorer fit [99]. Section 4.1.5 shows the sensitivity study for the time nodes, for both the node spacing and the maximum time to measure delayed neutrons.

Another consideration for the non-linear least squares solve is the covariance, which is larger for the pulse irradiation fit than for the saturation irradiation fit. This results from multiplying the group yield by the group decay constant. Attempts to replace this multiplication with an independent term produce poor fits, as the multiplication is not independent and is necessary for accurate modeling.

The non-linear least squares process also involves linearization and normalization of the data. The normalization is handled by taking the delayed neutron count rate, dividing by the fission term ¹, and taking the natural logarithm. The normalized count rate is then passed into the non-linear least squares fitting. The linearization is handled by taking the natural logarithm of the delayed neutron count rate. This treatment reduces both the covariance and the χ^2 of the fit. Bounds from 0 to 1000 are set for each term to speed up the process, while convergence tolerances are set slightly above machine tolerance at 2.23×10^{-16} to ensure a well-refined solution. The convergence criteria include changes in the independent variables and the norm of the gradient, with termination occurring when either criterion meets the set tolerance.

Sample self-multiplication, mentioned in Section 2.1.1, is another consideration. However, since the fission counts are taken directly from OpenMC tallies, self-multiplication is included and does not introduce error.

The final consideration in DNP group reconstruction is uncertainty. This work tracks uncertainties using the uncertainties package [41]. These uncertainties include uncertainties in cross sections, fission yields, half-lives, and emission probabilities. These uncertainties are used in the χ^2 minimization, as shown in Equation (2.6).

Once the group fits and their uncertainties are generated, the final step to obtain the finalized group parameters is to combine the saturation and pulse irradiation results. Saturation irradiation provides better statistics for longer-lived groups, while pulse irradiation is better for shorter-lived groups. Brady combines the results by using four long-period groups from saturation irradiation and four short-period groups from pulse irradiation. The groups are renormalized at the second group, where the two longest-lived group half-lives and yield ratios use the saturation results [21]. The remaining groups use the pulse irradiation results.

This work uses a similar but slightly simplified approach. The group parameters are first calculated for both pulse and saturation irradiation. Then, the two longest-lived groups use the saturation irradiation results, while the four shortest-lived groups use the pulse irradiation results.

3.5 Impact of DNP Groups on Transients

The impact of the DNP groups on transients is evaluated using Moltres with fluid-flow-velocity field coupled multi-group diffusion kinetics, as discussed in Section 2.1.6. This approach provides high-fidelity results by coupling thermal-hydraulics and neutronics, a necessity for MSR modeling. Modeling every DNP in space and time would be prohibitively expensive, so they are instead modeled using DNP group parameters. Moltres allows direct comparison between static DNP group parameters, which do not account for MSR phenomena, and parameters that do.

The results of interest from this analysis include the reactivity and power responses to various initiating events. The initiating events and transients vary, as different reactor responses are studied. The events of interest depend on the perspective from which the reactor response is analyzed.

Specifically, there are four different perspectives of interest in this work: validation, safety, security, and safeguards. Each perspective involves a unique combination of motivation, proposed initiating event, and investigated response. These are discussed in detail in their respective sections.

¹The fission term is either the net number of fissions or the fission rate, depending on the irradiation type.

3.5.1 Validation

The motivation for validation is to confirm that the proposed methodology and results align with experimental data. This ensures the demonstrated updated DNP group parameters are applicable to actual MSRs. Discrepancies between experimental results and the model highlight potential weaknesses in the approach. If no discrepancies exist, confidence in the results increases for problems similar to those validated.

The validation of the DNP group parameters follows a multi-stage approach. The first stage compares the irradiated sample's delayed neutron count rate to the Keepin et al. experiment from 1957 [66]. If the count rates do not match, the issue lies in the data or the aggregation of individual DNPs into the count rate. If the count rate is accurately predicted, the formulation is considered valid. The second stage validates the generated DNP group parameters using MSRE pump start-up and coast-down experimental data to assess their accuracy in modeling reactivity changes from DNPs [109]. This experiment has been studied using Moltres and Quasimolto, allowing comparison with their results [112, 101]. However, previous models excluded the inlet and outlet plena, leading to discrepancies with experimental data. A model including the plena, combined with updated DNP group parameters, should yield results closer to the experimental data.

3.5.2 Safety

The motivation for studying safety is to protect the public and meet regulatory requirements. Safety is discussed in detail in Section 2.3.1, though the general takeaway is that many safety-related transients exist. The DNP group parameters are critical in these transients, as they influence the reactor period, power, and temperature.

The transients investigated are ULOHS, ULOF, UTOP, and FSOC. For each transient, the reactor period, power response, maximum temperature, minimum temperature, and reactivity response are recorded. The reactor period is critical, as it reflects the reactor's controllability. The power response shows how changes in the DNP group parameters affect other responses, such as temperature. It is important to make sure differences in the DNP parameters don't lead to temperature changes or oscillations which result in unsafe reactor conditions.

3.5.3 Security

Security is discussed in detail in Section 2.3.2, but the general takeaway is that bad actors may attempt to sabotage the reactor. This ties directly into the safety design study, as sabotage aims to create unsafe reactor conditions. Ensuring the MSR operates safely after sabotage is critical. If static DNP group parameters predict safe operation while updated parameters predict unsafe operation, correcting this discrepancy in the reactor design is essential.

Since the safety study covers most transients relevant to sabotage, additional considerations can be incorporated into these transients. As sabotage may initiate multiple safety-related transients simultaneously, it is reasonable to investigate such scenarios. With four unique transients, this results in six double-transient combinations, three triple-transient combinations, and one quadruple-transient combination, totaling ten security-related sabotage-initiated transients.

3.5.4 Safeguards

Safeguards are discussed in Section 2.3.3, but the main takeaway is the importance of preventing SNM diversion in MSRs. Detection methods, such as neutron detectors and surveillance systems, play a key role in safeguarding the reactor [154]. The updated DNP group parameters in this work affect the effective delayed neutron fraction and reactivity response. Using static DNP group parameters to detect SNM diversion predicts less accurate reactivity

responses than the updated parameters. If the difference is significant, relying on static parameters may reduce the detection probability of a diversion scenario.

To investigate this, diversion scenarios are studied with a single significant quantity of material removed over the course of one year, simulated using OpenMC. The DNP group parameters are then generated for the fuel composition at that time, weighted by the volumetric fission rate fraction of each fissile nuclide, F , similar to Equation (2.10). For example, the total delayed neutron yield is given as:

$$\bar{v}_d = \left(\sum_F \int \Sigma_f^{(F)} \phi \, dE d\vec{r} \right)^{-1} \sum_F \left(\bar{v}_d^{(F)} \int \Sigma_f^{(F)} \phi \, dE d\vec{r} \right), \quad (3.10)$$

where,

$$\bar{v}_d^{(F)} = \text{the total delayed neutron yield for nuclide } F \text{ [-]} \quad (3.11)$$

$$\Sigma_f^{(F)} = \text{the macroscopic fission cross section for nuclide } F \text{ [cm}^{-1}\text{]}. \quad (3.12)$$

These weighted group parameters are generated using the static and the updated values. Then, they can be used in simple step reactivity insertion transients, where the following four sets of data are compared: static group parameters without diversion, static group parameters with diversion, updated group parameters without diversion, and updated group parameters with diversion. If the difference between the static group parameters with and without diversion is smaller or similar to the difference between the static and updated group parameters, then this means detection of that diversion scenario requires models to account for the MSR physical phenomena when generating group parameters.

Chapter 4

Preliminary Results

4.1 Sensitivity Studies

These sensitivity studies are used to analyze parameters that are held constant during the simulations and determine if simplifications can be made to some with negligible impact on the results. For these simulations, the parameters, unless otherwise specified, are given in Table 4.1.

Parameter	Value	Units
OpenMC Batches	10	-
OpenMC Particles	50000	-
OpenMC Depletion Chain	ENDF/B-VIII.0	-
OpenMC Cross Sections	ENDF/B-VIII.0	-
OpenMC Source Particles	4.2×10^{16}	-
Temperature	920	K
Neutron Energy	0.0253	eV
Sample Nuclide	^{235}U	-
Sample Density	10	$\frac{g}{cm^3}$
In-core Residence Time	20	s
Ex-core Residence Time	0	s
Irradiation Time	420	s
Decay Time	420	s
Decay Time Step	1	s

Table 4.1: Default parameters used for sensitivity studies of constant parameters used in 0D flow DNP group parameter formulation OpenMC simulations.

OpenMC batches and OpenMC particles both refer to the settings in OpenMC which dictate the number of batches per generation and the number of particles per batch, respectively. The source particles parameter dictates the strength of the neutron source. The number of source particles is divided by the irradiation time to generate the neutron source rate used in OpenMC during the irradiation, as shown in Equation (3.6). The irradiation time is the amount of time the sample is irradiated, the decay time is the amount of time the sample has the delayed neutron count rate measured after irradiation, and the decay time step is the time step used when updating the nuclide

concentrations post-irradiation. Additionally, by default, no chemical removal is modeled.

4.1.1 OpenMC Particles

If this value can be reduced, then the total amount of simulation time can be reduced. Alternatively, the default value may be insufficient for this problem and lead to inaccurate results. The model is exceedingly simple, composed of only a small sphere of fissile material and a vacuum boundary condition. Table 4.2 shows the total delayed neutron yields and average half-life for varying numbers of simulated particles in OpenMC. Because more particles lead to a slower simulation, a balance must be found between accuracy and cost. The differences between the largest number of particles for the total delayed neutron yield, $\bar{v}_d$, and the average half-life, $\bar{T}$, are given in the final two columns of the table. Based on these values, 50000 particles results in a fit accurate for the average half-life and total yield while also being somewhat computationally inexpensive.

Particles	$\bar{v}_d$	$\bar{T}$ [s]	$ \Delta \bar{v}_d $	$ \Delta \bar{T} $ [s]
1	0.016339	7.343276	0.002394	0.028612
10	0.019469	7.313167	0.000736	0.001497
100	0.018841	7.314828	0.000108	0.000164
500	0.018810	7.314185	0.000077	0.000478
1000	0.018693	7.315162	0.000040	0.000499
5000	0.018667	7.314440	0.000066	0.000223
10000	0.018700	7.314947	0.000033	0.000284
50000	0.018741	7.314586	0.000008	0.000077
100000	0.018730	7.314608	0.000002	0.000055
500000	0.018731	7.314636	0.000002	0.000028
1000000	0.018733	7.314663	-	-

Table 4.2: Sensitivity study of total delayed neutron yields and average half-lives due to number of particles simulated in OpenMC.

The variation in these values for lower particle counts is driven by differences in the group 2, 3, and 4 fits. Figure 4.1 shows the group yields for each different number of particles, where the largest differences are in groups 2, 3, and 4. These group half-lives range from 4.4 to 24 seconds. The cause of this difference must be from differences in nuclide concentrations post-irradiation, as the emission probability and decay constant data are the same between the different simulations compared here. To evaluate which nuclides are causing this difference, the standard deviation of each set of concentrations is divided by the mean, which results in the coefficient of variation (CV) for each nuclide. Some of the nuclides with the largest CV have a negligible contribution to the delayed neutron yield, such as ^{209}Tl . To scale by this consideration, the CV is multiplied by the total number of delayed neutrons emitted from that nuclide i , $\bar{v}_{d,i}$, the equation for which is given in Equation (2.21), replicated here:

$$\bar{v}_{d,i} = P_{n,i} \lambda_i N_{i,0}$$

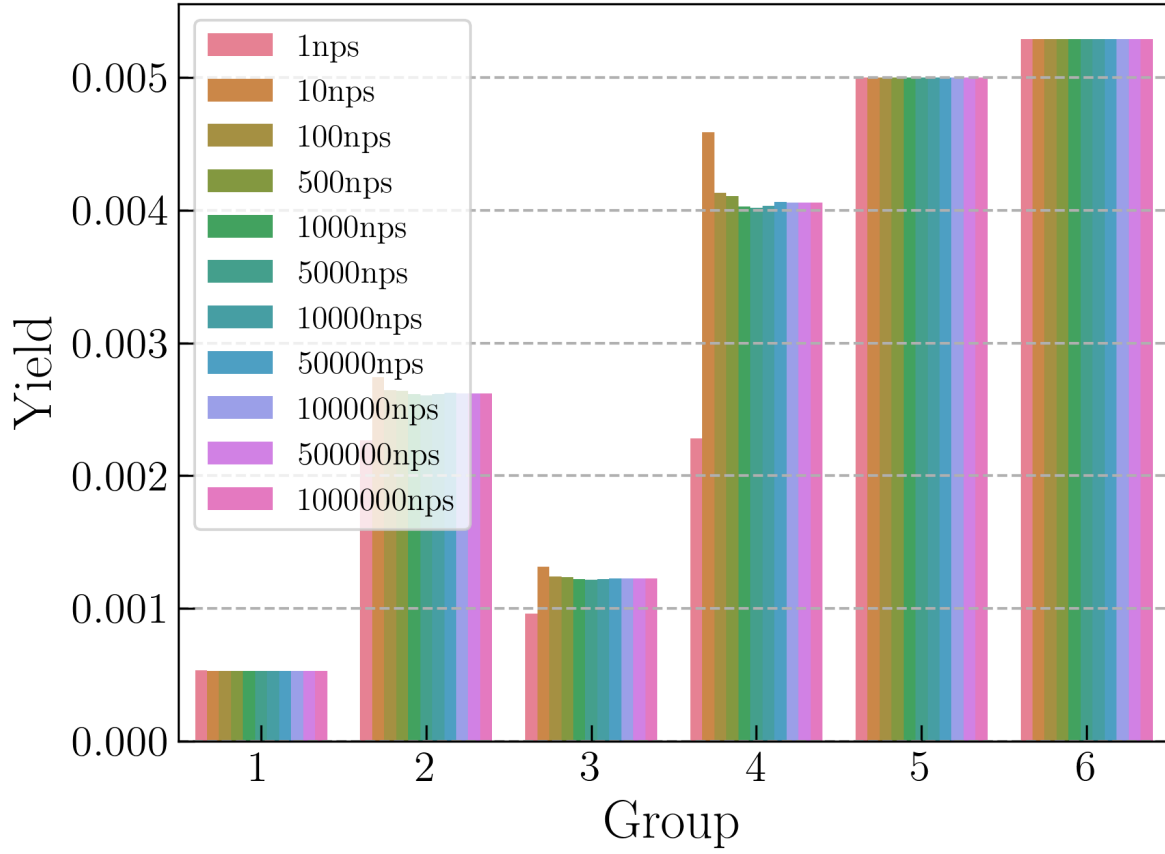


Figure 4.1: Group yields resulting from various numbers of simulated particles in OpenMC.

The scaling of the CV by the total delayed neutron yield is important because nuclides like ^{209}Tl have a CV of 9.65%, but a total delayed neutron yield of 0. The nuclides are sorted by a new scoring term, α , where the full term in Equation (4.1) is given as:

$$\alpha_i = \frac{\bar{v}_{d,i} CV}{\bar{v}_d} \quad (4.1)$$

Based on α , only two of the top ten nuclides are within groups two through four, with the nuclides being ^{89}Br and ^{137}I . When sorting by CV instead, none of the top thirty nuclides are within groups two through four. If the CV is replaced with the standard deviation, then there are six out of ten nuclides which are within groups two through four. These nuclides are ^{137}I , ^{88}Br , ^{93}Rb , ^{89}Br , ^{138}I , and ^{136}Te . Although the relative changes in these nuclides are smaller, the absolute change is larger, which drives a greater difference in groups 2 through 4.

4.1.2 Temperature

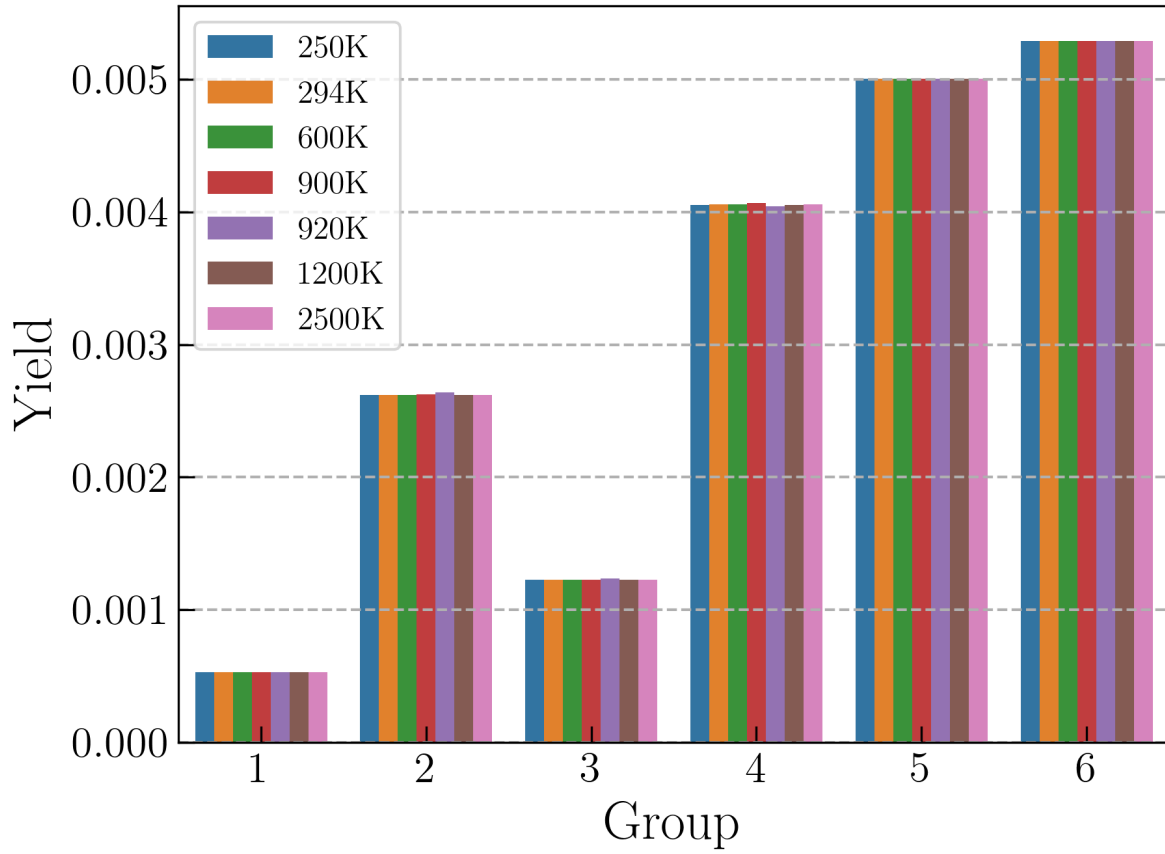


Figure 4.2: Group yields resulting from various sample temperatures in OpenMC.

Temperature [K]	$\bar{\nu}_d$	$\bar{T}$ [s]
250	0.018729	7.315157
294	0.018732	7.314929
600	0.018735	7.314949
900	0.018743	7.314833
920	0.018745	7.318251
1200	0.018722	7.314862
2500	0.018732	7.315089

Table 4.3: Sensitivity study of total delayed neutron yields and average half-lives due to the sample temperature simulated in OpenMC.

4.1.3 Target Density

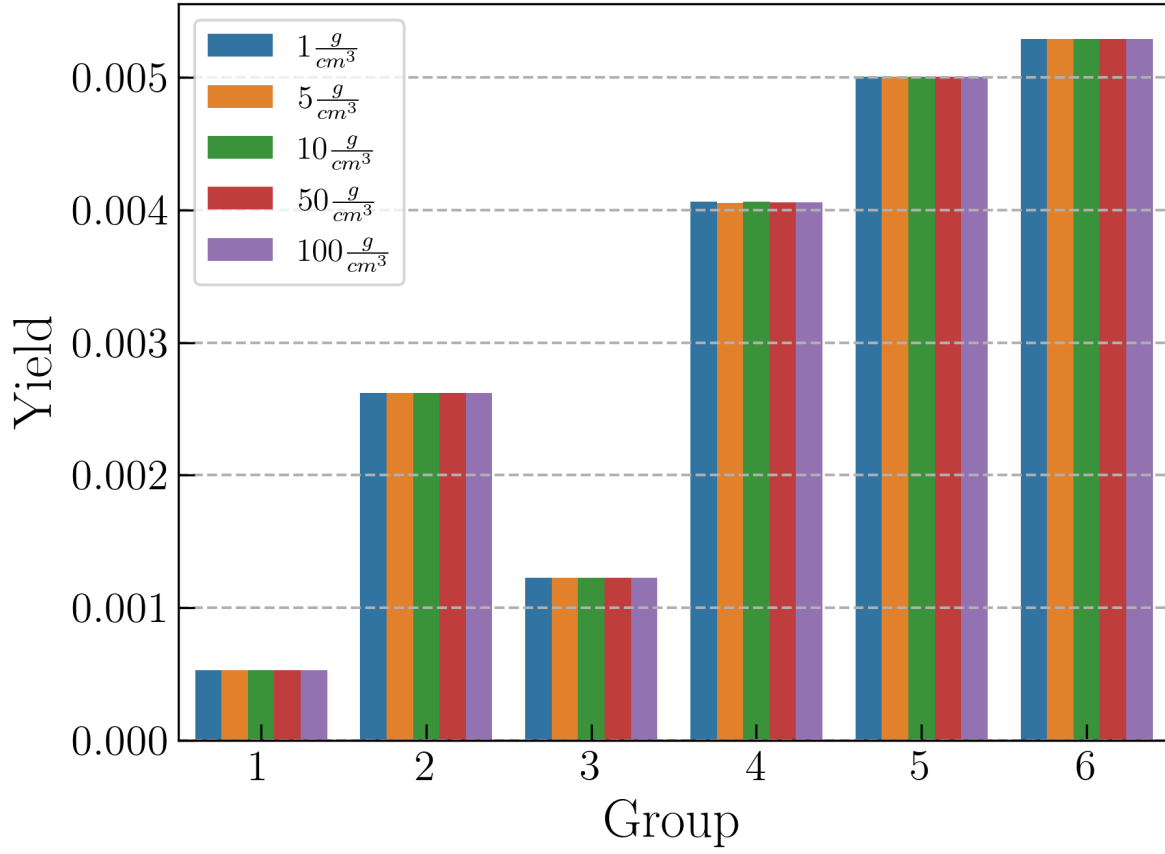


Figure 4.3: Group yields resulting from various sample densities in OpenMC.

Density [$\frac{g}{cm^3}$]	$\bar{\nu}_d$	$\bar{T}$ [s]
1	0.018738	7.313331
5	0.018725	7.313464
10	0.018735	7.313371
50	0.018731	7.313418
100	0.018730	7.313413

Table 4.4: Sensitivity study of total delayed neutron yields and average half-lives due to sample density simulated in OpenMC.

4.1.4 Irradiation Time

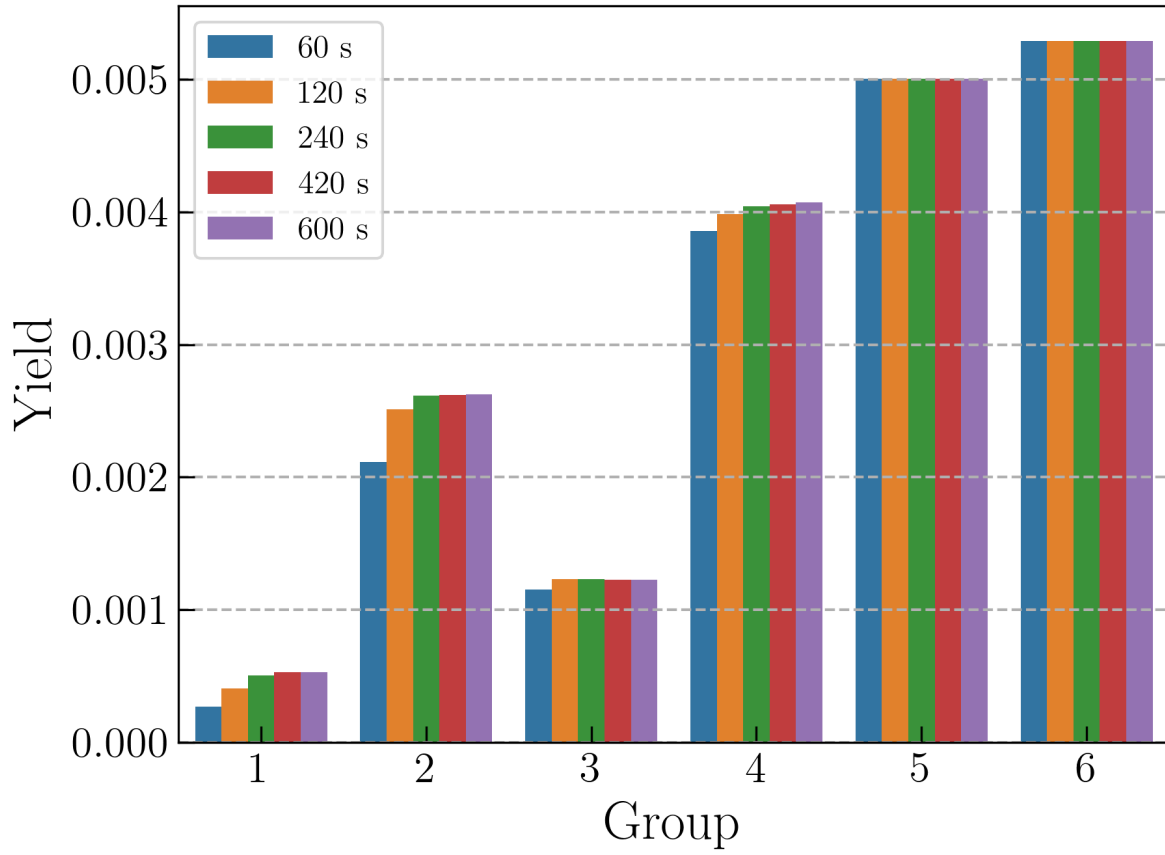


Figure 4.4: Group yields resulting from various irradiation times in OpenMC.

Time [s]	$\bar{\nu}_d$	$\bar{T}$ [s]	$ \Delta \bar{\nu}_d $	$ \Delta \bar{T} $ [s]
60	0.017690	7.388026	0.001062	0.066482
120	0.018433	7.355979	0.000319	0.034434
240	0.018687	7.329571	0.000065	0.008026
420	0.018731	7.322325	0.000021	0.000780
600	0.018752	7.321545	-	-

Table 4.5: Sensitivity study of total delayed neutron yields and average half-lives due to net irradiation time simulated in OpenMC.

4.1.5 Decay Time and Decay Time Steps

The bulk of the computational cost comes from writing statepoint files; sufficient resolution is required to ensure a smooth decay curve.

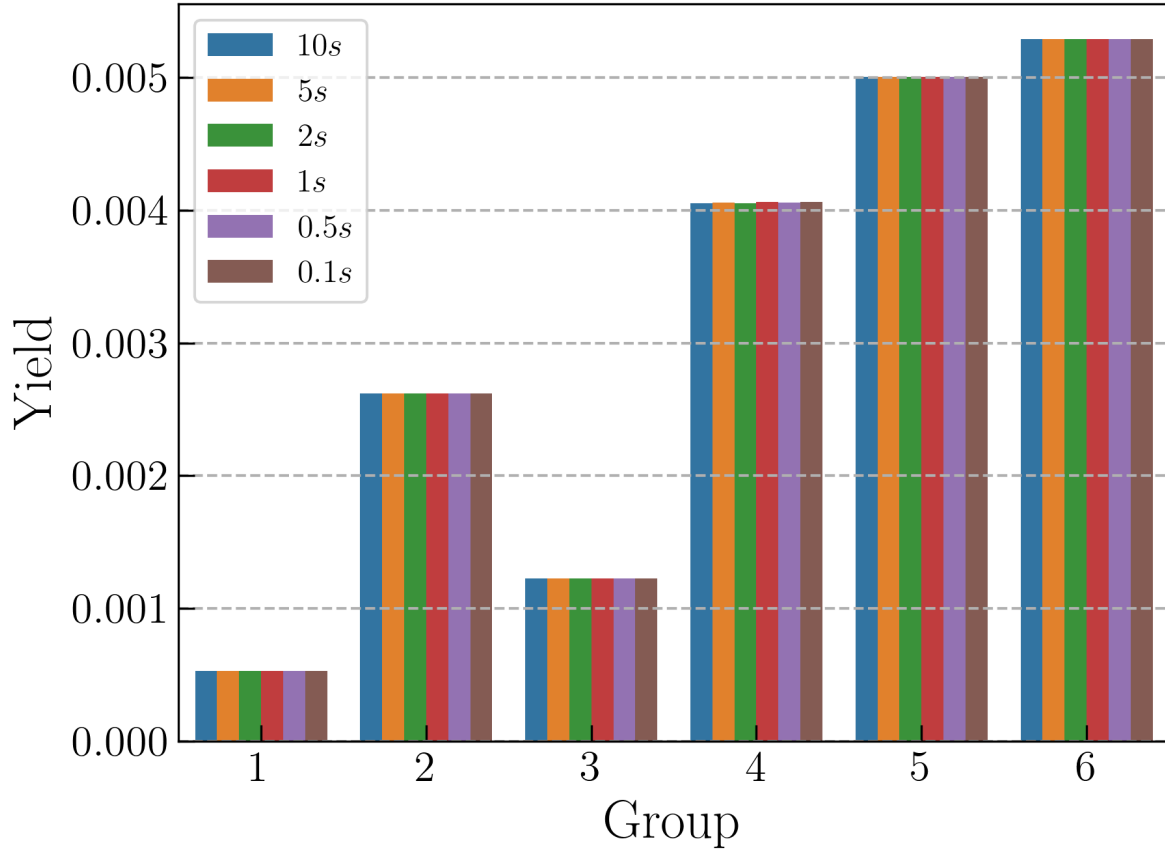


Figure 4.5: Group yields resulting from various decay time steps in OpenMC.

Δt [s]	$\bar{\nu}_d$	$\bar{T}$ [s]	$ \Delta \bar{\nu}_d $	$ \Delta \bar{T} $ [s]
10s	0.018722	7.314094	0.000013	0.000076
5s	0.018732	7.314083	0.000003	0.000065
2s	0.018729	7.314162	0.000006	0.000144
1s	0.018735	7.314021	0.000000	0.000003
0.5s	0.018731	7.314008	0.000004	0.000010
0.1s	0.018735	7.314018	0.000000	0.000000

Table 4.6: Sensitivity study of total delayed neutron yields and average half-lives due to decay time step simulated in OpenMC.

Using a time step of 1 second:

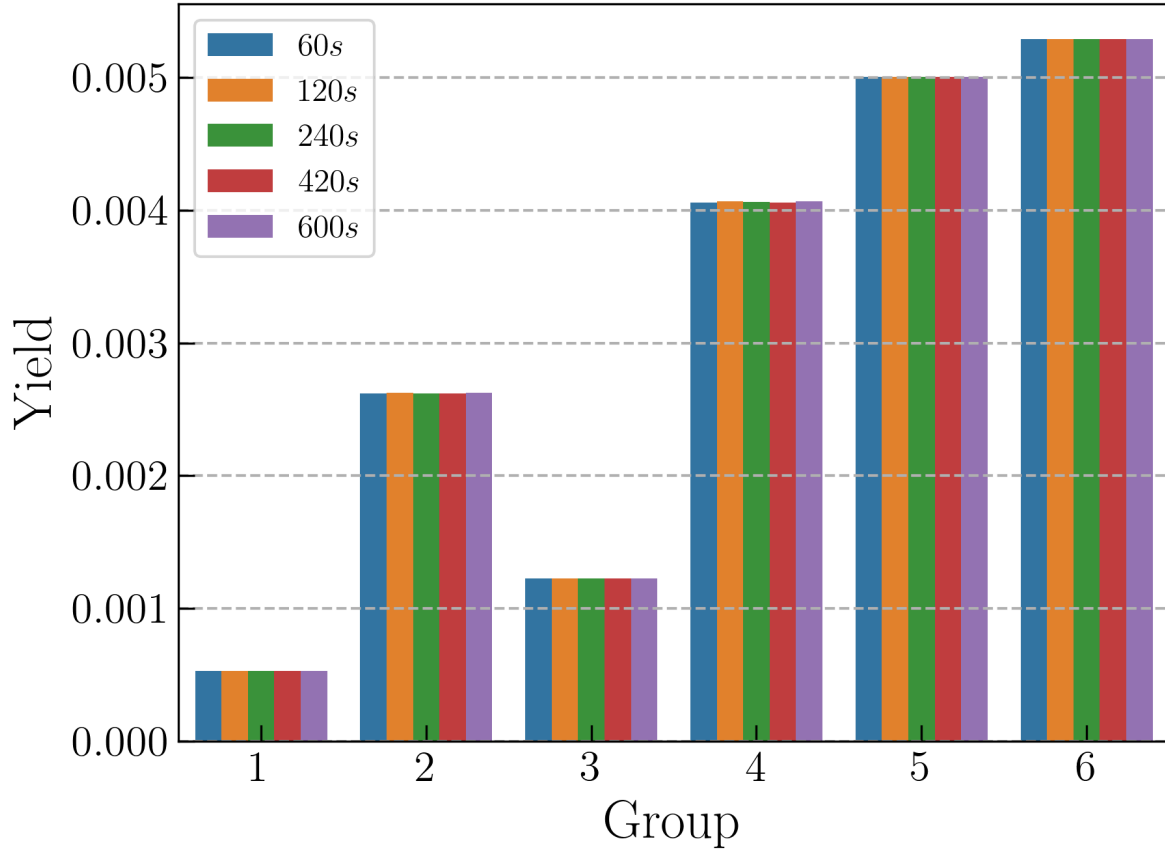


Figure 4.6: Group yields resulting from various decay times in OpenMC.

t [s]	$\bar{\nu}_d$	$\bar{T}$ [s]	$ \Delta \bar{\nu}_d $	$ \Delta \bar{T} $ [s]
60s	0.018734	7.315858	0.000010	0.000039
120s	0.018744	7.315819	0.000000	0.000000
240s	0.018735	7.315828	0.000009	0.000009
420s	0.018734	7.315880	0.000010	0.000061
600s	0.018744	7.315819	0.000000	0.000000

Table 4.7: Sensitivity study of total delayed neutron yields and average half-lives due to decay time simulated in OpenMC.

4.1.6 Large Cycle Time Elements

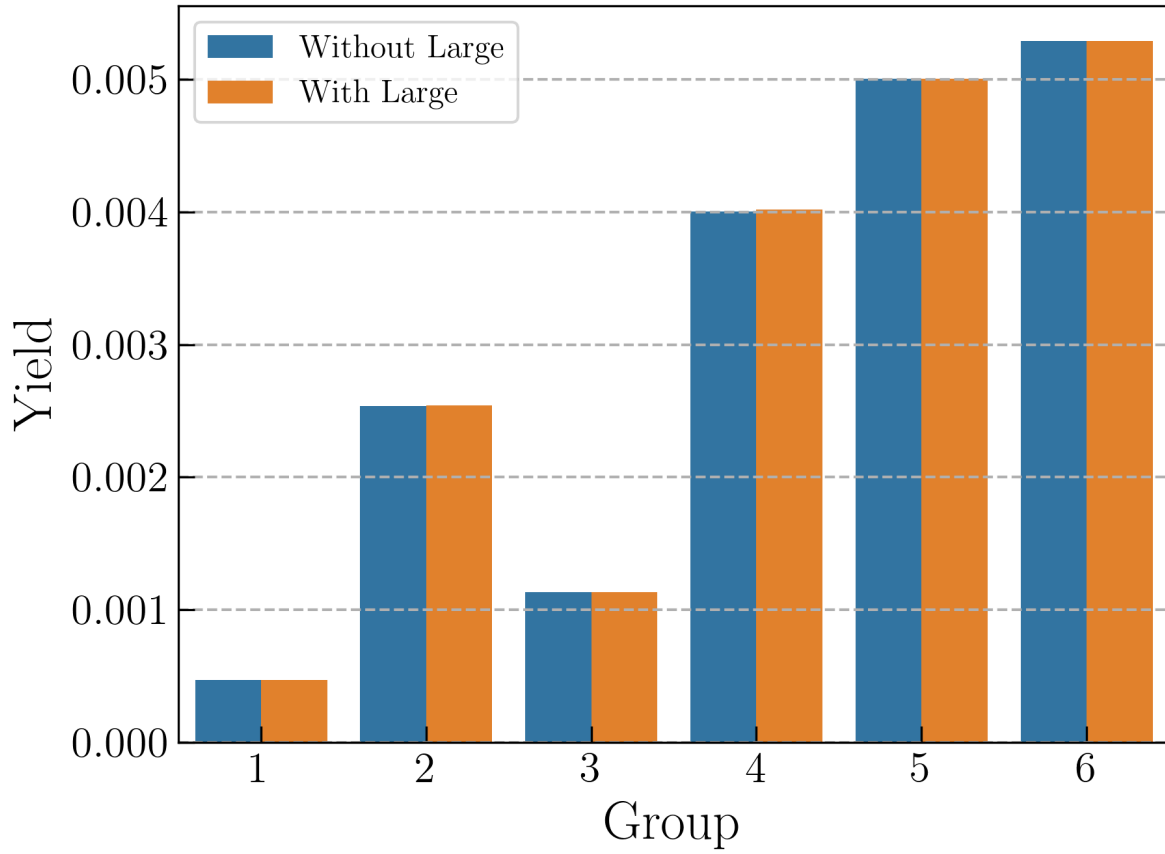


Figure 4.7: Group yields with and without large cycle time reprocessing in OpenMC.

Large Cycle Time Reprocessing	$\bar{\nu}_d$	$\bar{T}$ [s]	$ \Delta \bar{\nu}_d $	$ \Delta \bar{T} $ [s]
False	0.018442	7.060040	0.000018	0.000029
True	0.018460	7.060011	-	-

Table 4.8: Sensitivity study of total delayed neutron yields and average half-lives due with and without large cycle time reprocessing simulated in OpenMC.

4.1.7 Decay Daughter Tracking

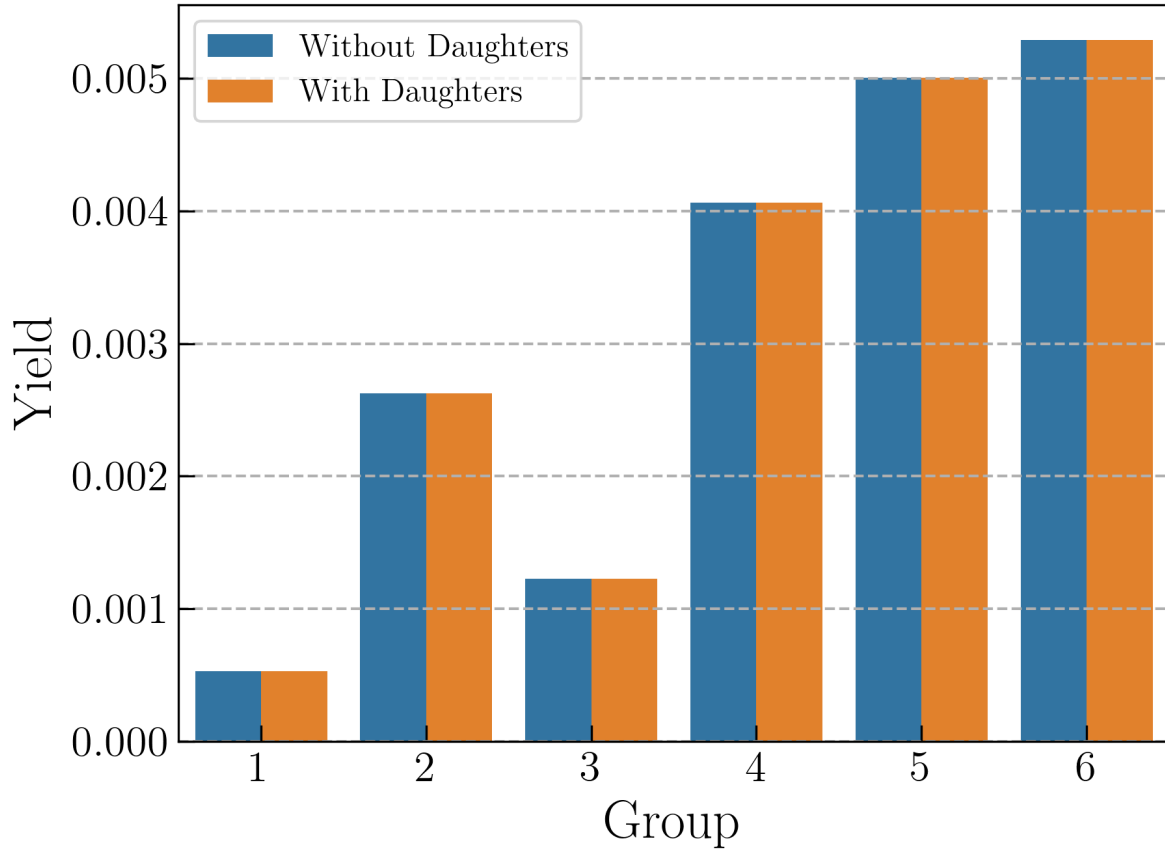


Figure 4.8: Group yields with and without decay chains simulated after irradiation in OpenMC.

Decay Daughters Modeled	$\bar{\nu}_d$	$\bar{T}$ [s]	$ \Delta \bar{\nu}_d $	$ \Delta \bar{T} $ [s]
False	0.018740	7.315761	0.000001	0.000109
True	0.018741	7.315652	0.000000	0.000000

Table 4.9: Sensitivity study of total delayed neutron yields and average half-lives due with and without decay chains simulated after irradiation in OpenMC.

4.1.8 Takeaways

50k particles (0.8 pcm difference very small)

Decay time step of 1 second (becomes more expensive for more time steps) Irradiation time of 600 seconds (difference of 2.1 pcm small, but might as well account for it) Large cycle time elements have 1.8 pcm difference, but negligible computational cost.

All these small differences add up. Do a comparison between the full fidelity results and results with +/- 2 pcm.

Decay daughters don't need to be tracked, but cost is small (many depletion simulations, but they proceed rapidly).

4.2 Verification of 0D Flow Model

[81] table 2 has group parameters, these can be used as a direct comparison and the effects can be compared for the MSFR.

[90] has MSFR PRK and IPK which can be compared.

fluid flow velocity change on power (Dulla PhD thesis I can't access)

[76] - can compare reactivity insertion effects on power

[90] - extracted power effect on power

[87] -

Microscopic group parameters [77, 93, 62] should match static results (no flow or chemical removal.)

ENDF Table II in [62] shows a yield of 1.9 for thermal u235 in ENDF, which matches the static results in the simplified model simulation.

4.3 Effects of MSR Phenomena on Group Parameters

These sensitivity studies are used to analyze the effects of the parameters that represent MSR physical phenomena. Polynomial fits of group parameters. Using MSRE temperature and salt density [117]. No reprocessing unless specified.

Parameter	Value	Units
OpenMC Batches	10	-
OpenMC Particles	50000	-
OpenMC Depletion Chain	ENDF/B-VIII.0	-
OpenMC Cross Sections	ENDF/B-VIII.0	-
OpenMC Source Particles	4.2×10^{16}	-
Temperature	920	K
Neutron Energy	0.0253	eV
Sample Nuclide	^{235}U	-
Sample Density	2.3275	$\frac{g}{cm^3}$
In-core Residence Time	600	s
Ex-core Residence Time	-	s
Irradiation Time	600	s
Decay Time	600	s
Decay Time Step	1	s

Table 4.10: Default parameters used for study on MSR phenomena in 0D flow DNP group parameter formulation OpenMC simulations.

Removal Rates

In and Ex-core Residence Times

Compare with static. MSRE is approximately 10 seconds in-core and 15 seconds ex-core [117].

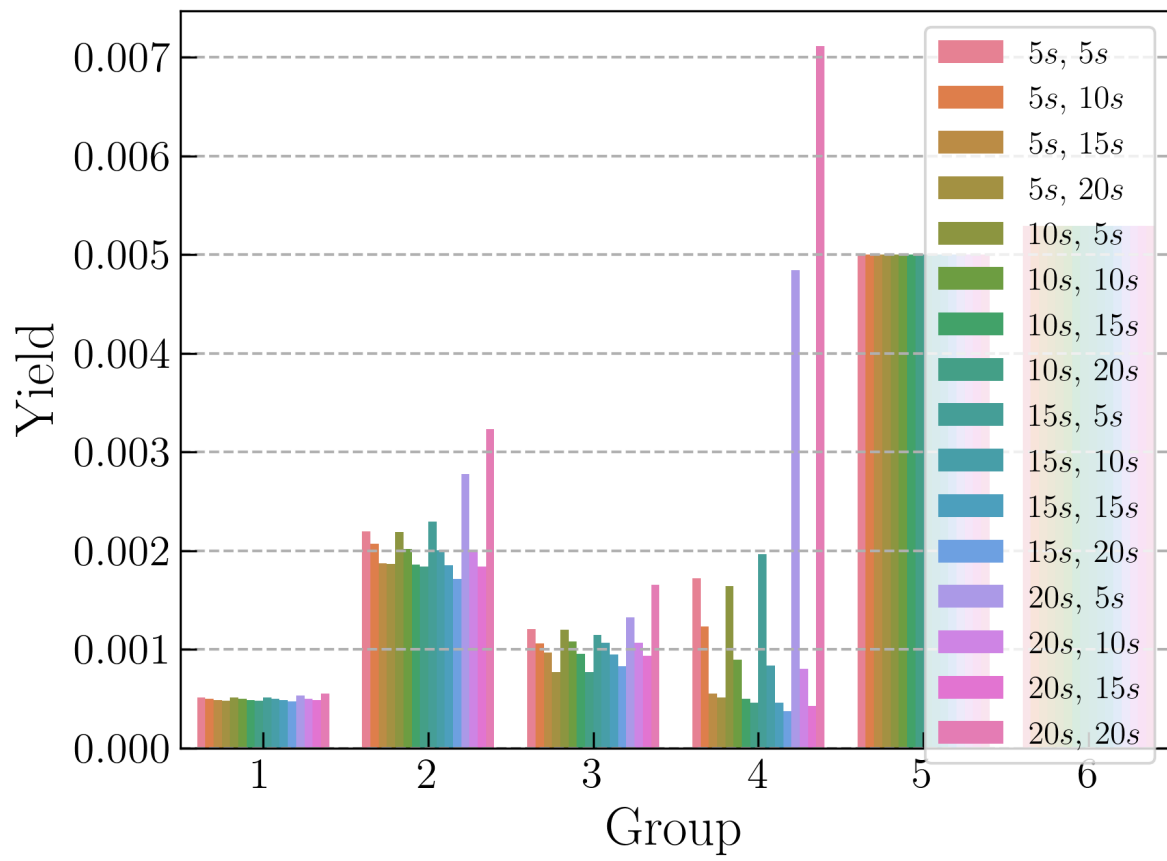


Figure 4.9: Group yields for various residence times during irradiation in OpenMC.

τ_{in} [s]	τ_{ex} [s]	$\bar{v}_d$	$\bar{T}$ [s]
5s	5s	0.015940	8.101997
5s	10s	0.015165	8.030291
5s	15s	0.014179	8.293764
5s	20s	0.013929	7.770002
10s	5s	0.015851	8.113214
10s	10s	0.014790	8.234031
10s	15s	0.014109	8.304067
10s	20s	0.013845	7.819340
15s	5s	0.016220	7.869896
15s	10s	0.014718	8.240623
15s	15s	0.014051	8.310107
15s	20s	0.013693	8.191821
20s	5s	0.019775	7.537696
20s	10s	0.014672	8.244369
20s	15s	0.013996	8.313858
20s	20s	0.022849	7.536158

Table 4.11: Effect of residence times on the total delayed neutron yield and average half-life.

Combined Removal Rates and Residence Times

Chapter 5

Proposed Work

5.1 Group Parameter Validation

The validation of the DNP group parameters is important to ensure the results will be useful in practice. Although the group parameters are verified with other group parameters in the literature, this does not necessarily mean they are capturing all of the physical phenomena of interest. Validation shows where the data may be lacking and conveys potential weaknesses of the approach, depending on the discrepancies between the predicted results and actual results.

Validation of the group parameters generated in this work is conducted in two stages, as discussed in Section 3.5. The first stage validates the delayed neutron count rate formulation and group generation for a static problem. The second stage validates the updated DNP group parameters by comparing the reactivity feedback with that of the static group parameters.

The first stage is validation of the irradiation by comparing the irradiated sample's generated delayed neutron count rate to that of the Keepin et al. experiment from 1957 [66]. This requires modeling Godiva in OpenMC and solving the k-eigenvalue problem to irradiate the sample. The concentration of the sample after irradiation is then passed to the scripts, which will generate the DNP group parameters. The delayed neutron count rate and group parameters are compared with the values from Keepin et al. to determine if there are discrepancies. Some difference is expected due to uncertainties in the data and the experimental results, so the validation can be considered successful provided the results align within a few percent. If they do not align, then that would indicate that there is either a bug which was not present during verification or an issue with the data. If there is an issue with the data, it will be noted for additional future work to improve the microscopic data so that the methodology proposed in this work can be more successfully implemented in the future.

The second stage of the validation is for the generated DNP group parameters. This stage uses the MSRE pump start-up and coast-down experimental data to see if the group parameters accurately model the reactivity changes from the DNPs [109]. This stage requires modeling the MSRE core and upper and lower plena in Moltres. The effect of the DNPs on reactivity is then calculated for both transients using both static and updated DNP group parameters. The expected result is that the updated group parameters will more accurately predict the reactivity from the DNPs due to the change in the flow rate. If neither set of group parameters aligns with the experimental results, then that would indicate an issue with the model, since it has been shown previously that Moltres can model these transients [112, 101]. If the static group parameters more accurately predict the reactivity, then that would indicate that there is an issue with the formation of the updated group parameters, particularly if the first stage of

Data name	Year	Source	Half-life	Emission probability	Concentration
IAEA database	2018	[37]	Yes	Yes	No
JEFF 3.3	2017	[?]]	Yes	Yes	Yes
ENDF/B-VIII.0	2018	[23]	Yes	Yes	Yes
JENDL-5	2021	[?]]	Yes	Yes	Yes
ENSDF	2025	[?]]	Yes	Yes	No
Audi et al.	2003	[7]	Yes	Yes	No
Pfeiffer and Kratz	2002	[106]	Yes	Yes	No
Rudstam et al.	1993	[122]	Yes	Yes	No

Table 5.1: Various data sources as well as the data which they contain.

validation was successful. The issue will have to be resolved before the work can continue, since the updated DNP group parameters should provide higher accuracy relative to the static group parameters in MSR modeling.

5.2 Data Evaluation

The use and handling of the various data has a direct impact on the DNP group parameters. These impacts can be caused by the data source, time resolution of the fission yield, number of DNPs included in the model, and energy of the incident neutrons used to irradiate the sample. The data source includes the source of cross-section data, fission yields, emission probabilities, half-lives, and other associated data used to generate the DNP group parameters. The time resolution of the fission yield is the difference between using a cumulative fission yield and a time-resolved depletion chain for evaluating DNP concentrations. The number of DNPs is important because modeling more DNPs is more computationally expensive for the spatially resolved model, so minimizing this term without affect DNP group generation is of interest. Finally, the incident neutron energy spectra is used to study the most effective way of organizing the DNP group parameters with regards to energy, such as using the median or mean energy.

5.2.1 Data Sources

Huynh et al. compare the effects of cumulative fission yield and emission probabilities from the Joint Evaluated Fission and Fusion (JEFF), ENDF, and Japanese Evaluated Nuclear Data Library (JENDL) on the total delayed neutron yield [62]. Mathieu et al. investigate different datasets as well, including the National Nuclear Data Center (NNDC) website (the data in which comes from Evaluated Nuclear Structure Data File (ENSDF)) and various data compilations [87, 106, 7, 122]. This can be expanded upon by comparing the IAEA database, NNDC website, JEFF, ENDF, JENDL, ENSDF, and various data compilations for various combinations of emission probabilities (also called delayed neutron branching ratios), half-lives, fission yields, and cross sections. By comparing these different data sources, the uncertainty caused by differences in the data can be more clearly understood. Table 5.1 shows the selection of data sources which will be used in the comparison. The "Concentration" column refers to data such as fission yields and cross-sections which affect the DNP concentration term. Although the half-life of the nuclides also affects the concentration, the data sources which do not have a concentration contribution only provide relevant data on the half-lives and emission probabilities. Audi et al., Pfeiffer and Kratz, and Rudstam et al. are presentations of data from the literature rather than official databases. Audi et al. use ENSDF, Atomic Mass Evaluation (AME), experimental data from the literature, and trends in neighboring nuclides. Pfeiffer and Kratz present experimental data, data from an update to the Kratz Herrmann formula (KHF), and data from a unified macroscopic-microscopic model within the quasi-particle random-phase approximation (QRPA). Rudstam et al. presents experimental data.

The expected outcome of using these varying data sources is the following: more recently updated data will be more accurate and the official databases will be more accurate than the collections from the literature. The more recent data should be more accurate, as new experimental data should be an improvement due to better equipment and measurement techniques. The official databases are expected to be more accurate, as they have many users that can report mistakes, and they have been developed for an extended period of time. It will be useful to investigate these data sources and determine the differences between the DNP group parameters generated for each. If the differences are large, then it will be important to understand in regards to propagation of the uncertainty, while negligible differences will simplify further uncertainty analysis.

5.2.2 Fission Yield Time Resolution

A relevant analysis to be conducted is comparing the cumulative fission yield to the full depletion approach. Previous microscopic DNP group parameter evaluations have used cumulative fission yields rather than depletion simulations. This approach does not capture the full time dependence of decay chains leading to DNPs, which could result in discrepancies when including the MSR physical phenomena. However, because using the cumulative fission yields would remove the need for depletion simulations, it is useful to determine whether the cumulative fission yield approach can be used. It is also useful because the spatially resolved method is much more efficient using the cumulative fission yield rather than full spatially resolved depletion.

Because saturation irradiation, this should not matter for static. With flow and reprocessing, it could matter.

5.2.3 Number of DNPs

Another analysis of interest in regards to the data is on the number of DNPs needed in the spatially resolved method to generate accurate results. The computational cost and memory requirements for the spatially resolved method scale with the number of nuclides included, so minimizing them is of interest. The impact of each DNP is given in Equation (2.21), replicated here:

$$\tilde{v}_{d,i} = P_{n,i} \lambda_i N_{i,0}.$$

Dividing $\tilde{v}_{d,i}$ by the total delayed neutron yield, $\tilde{v}_d$, provides the relative contribution of each DNP to the total delayed neutron yield. The DNPs with the most impact are modeled preferentially to determine the minimum number of DNPs to model with negligible changes to the DNP group parameters.

5.2.4 Incident Neutron Energy Spectra

Analyzing the effect of a neutron irradiation spectrum compared to a single neutron irradiation energy is also of interest. Particularly, the effect of various neutron irradiation spectra where the average and mean neutron irradiation energy matches that of a single energy irradiation.

5.3 Uncertainty Quantification

There are several sources of uncertainty which can be considered, including those from half-lives, emission probabilities, cross sections, and fission yields. These uncertainties propagate and results in uncertainty in the DNP group and kinetics parameters [111]. The uncertainties in the DNP group parameters are tracked through the non-linear least squares fitting by taking the square root of the generated covariance matrix. This uncertainty uses a linear approximation near the optimum [144].

There are also uncertainties in the MSR physical phenomena. These include uncertainties in the flow rate and reprocessing rates, where the reprocessing rate is correlated with the flow rate. The magnitude of this uncertainty is discussed in Section 4.3, where the sensitivity of the DNP group parameters to the changes in the MSR phenomena demonstrate the potential impact of uncertainties.

5.4 Generate DNP Group Spectra

The delayed neutron energy spectra, χ_d is important for accurate prediction of reactor kinetics [33]. This can be seen in the equation for the effective delayed neutron fraction in Equation (2.25), recreated here as:

$$\beta_{eff} = \frac{\int \Phi^\dagger \chi_d \beta \bar{\nu} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}{\int \Phi^\dagger \chi \bar{\nu} \Sigma_f \Phi dE d\vec{\Omega} d\vec{r}}.$$

The methodology to evaluate the delayed neutron spectra is discussed in Section 2.1.4.

5.5 Develop Spatially Resolved Model

The spatially resolved model is discussed in detail in Section 3.2.2. This spatially resolved model will be used in a comparison for multiple levels of fidelity of generating the DNP group parameters: data only using cumulative fission yields, OpenMC simulations with depletion, spatially resolved using cumulative fission yields, and spatially resolved with depletion. Each step up becomes more computationally expensive, so if a simpler method is able to generate DNP group parameters with negligible differences, it will be used preferentially.

The process for comparing the 0D flow model and the spatially resolved model is as follows: the spatially resolved model runs and generates DNP group parameters and residence times, the residence times are used to feed into the 0D flow model, the resulting DNP group parameters are compared between the two models. If the results match, then this would indicate that the 0D flow model only needs residence time data for a given MSR to generate the DNP group parameters. If they do not match, then a spatially resolved approach will be necessary to properly generate the MSR DNP group parameters.

5.6 Kinetics Simulations

The kinetics simulations are important for the validation of the DNP group parameters using the MSRE transients, as well as investigating transients which affect the security, safety, and safeguards design decisions. For safety design, the ULOF and UTOP are investigated for various reactor parameters to determine the effect of the updated DNP group parameters. For security design, the sabotage initiated transients are assumed to be the same as those studied for the safety design. For safeguards design, comparison is made between various diversion scenarios with different PR times and material quantities removed. These diversion scenarios affect the fuel composition, which affects the effective delayed neutron fraction, which will affect the behavior of the reactor during a transient. The updated DNP group parameters will be used to investigate the SNM removal which could remain undetected while using previous DNP group parameters.

5.7 Releasable Tool

The final proposed extension to this work is the release of an open-source Python package. This package will provide users with the capability to generate DNP group parameters for any MSR, with any fissile nuclide, energy spectra, and desired number of DNP groups. The plan is to offer a customizable and modular tool which is easily accessible, enabling users to study more effects of these updated DNP group parameters.

There are many additional effects to study which would be of interest, including the following: more refined data generation, investigation of other fissile nuclides and energy spectra, further optimizations to the methodology, and inclusion of more MSR physical phenomena such as thermochemistry. Other future work of interest would be experimental results replicating the methodology proposed in this work, specifically, irradiation of a fissile sample in a circulating loop and counting of the delayed neutrons from that sample.

Appendix A

Derivation Delayed Neutron Count Rate in 0D Flow for Equal Residence Times

This derivation begins with the Equations (AA.1) and (AA.2) provided by Wilson and England [146]:

$$R(t) = \sum_{k=1}^K \bar{v}_{d,k} \lambda_k e^{-\lambda_k t}, \quad (\text{AA.1})$$

and,

$$\dot{n}_d(T, t) = \bar{v}_d \int_0^T F(\tau) R(T - \tau + t) d\tau, \quad (\text{AA.2})$$

where,

T = the length of the fission history [s]

t = the decay time from the end of the fission history [s]

$F(\tau)$ = the fission rate history [$\frac{\text{fissions}}{\text{s}}$].

Because the results are normalized to a single delayed neutron, the $\bar{v}_d$ term is divided out.

To derive an analytical solution to this problem for the 0D flow approach where the in-core and ex-core residence times, τ_{in} and τ_{ex} , are equivalent, an approximation of the fission history must be made. For this derivation, τ_{res} will be used to represent both the in-core and ex-core residence times. Because the 0D flow model's fission history is a positive square wave, analytical integration is not feasible. Instead, a Fourier decomposition of the square wave into a sum of sine waves makes the problem far more approachable. This is given as:

$$\frac{F(t)}{F_0} = \frac{1}{2} + \frac{2}{\pi} \left(\sin\left(\frac{\pi t}{\tau_{res}}\right) + \frac{1}{3} \sin\left(\frac{3\pi t}{\tau_{res}}\right) + \dots \right), \quad (\text{AA.3})$$

where,

F_0 = the magnitude of the constant fission rate [$\frac{\text{fissions}}{\text{s}}$].

Plugging this in and separating the integral, for group k and for j Fourier terms, gives:

$$\begin{aligned}
n_{d,k}(T, t) = & \frac{F_0 \bar{v}_{d,k} \lambda_k}{2} \int_0^T e^{-\lambda_k(T-\tau+t)} d\tau \\
& + \frac{2F_0 \bar{v}_{d,k} \lambda_k}{\pi} \int_0^T \sin\left(\frac{\pi\tau}{\tau_{res}}\right) e^{-\lambda_k(T-\tau+t)} d\tau \\
& + \dots \\
& + \frac{2F_0 \bar{v}_{d,k} \lambda_k}{(2j+1)\pi} \int_0^T \sin\left(\frac{(2j+1)\pi\tau}{\tau_{res}}\right) e^{-\lambda_k(T-\tau+t)} d\tau
\end{aligned} \tag{AA.4}$$

Solving the integrals and simplifying yields:

$$\begin{aligned}
n_{d,k}(T, t) = & \frac{F_0 \bar{v}_{d,k} e^{-\lambda_k t} (1 - e^{-\lambda_k T})}{2} \\
& + \dots \\
& + \frac{2F_0 \bar{v}_{d,k} \lambda_k^2 \tau_{res}^2 e^{-\lambda_k t}}{((2j+1)\pi)((2\pi j + \pi)^2 + \lambda_k^2 \tau_{res}^2)} \sin\left(\frac{\pi(2j+1)T}{\tau_{res}}\right) \\
& + \frac{2F_0 \bar{v}_{d,k} \lambda_k \tau_{res} e^{-\lambda_k t} e^{-\lambda_k T}}{((2\pi j + \pi)^2 + \lambda_k^2 \tau_{res}^2)} \cos\left(\frac{\pi(2j+1)T}{\tau_{res}}\right) \\
& + \frac{2F_0 \bar{v}_{d,k} \lambda_k \tau_{res} e^{-\lambda_k t} e^{-\lambda_k T}}{((2j+1)\pi)((2\pi j + \pi)^2 + \lambda_k^2 \tau_{res}^2)} 2\pi j \\
& + \frac{2F_0 \bar{v}_{d,k} \lambda_k \tau_{res} e^{-\lambda_k t} e^{-\lambda_k T}}{((2j+1)\pi)((2\pi j + \pi)^2 + \lambda_k^2 \tau_{res}^2)} \pi,
\end{aligned} \tag{AA.5}$$

where each integrated Fourier term generates four terms added to the sum.

Similar to [146], the irradiation time is taken to the limit as it approaches infinity, though only for the non-periodic terms. By irradiating for several half-lives of time for each DNP, where the longest half-life is roughly one minute, then the irradiation time can be taken as infinite. There are two reasons this limit is taken. Firstly, a pulse irradiation with circulation will not provide any unique results when compared to traditional data which already exists. Additionally, this allows the equation to be simplified, as shown in Equation (AA.6). Only the sine term does not go to zero as the irradiation time goes to infinity, so it is the only term significant for saturation irradiation in the 0D flow model.

$$\begin{aligned}
n_{d,k}(T, t) = & \frac{F_0 \bar{v}_{d,k} e^{-\lambda_k t}}{2} \\
& + \dots \\
& + \frac{2F_0 \bar{v}_{d,k} \lambda_k^2 \tau_{res}^2 e^{-\lambda_k t}}{((2j+1)\pi)((2\pi j + \pi)^2 + \lambda_k^2 \tau_{res}^2)} \sin\left(\frac{\pi(2j+1)T}{\tau_{res}}\right)
\end{aligned} \tag{AA.6}$$

The final assumption to make is that the irradiation time, T , is an integer multiple of the residence times, τ_{res} . This assumption makes the sine term 0 for every value of k , which simplifies the result to:

$$n_d(t) = \frac{F_0}{2} \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t} \tag{AA.7}$$

This is very similar to the saturation irradiation delayed neutron count rate equation, which is:

$$n_d(t) = F_0 \sum_{k=1}^K \bar{v}_{d,k} e^{-\lambda_k t} \quad (\text{AA.8})$$

Because of the initial assumption made, which was that half the time is in-core and half ex-core, the average fission rate is half of the fission rate scaling term, F_0 . For both cases (constant irradiation and half in-core and half ex-core), the fission term in front is simply the average fission rate over the irradiation time.

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